06-Exercices-GARCH-VaR

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Exercice 1 : Modéliser le risque du CAC40

Données à partir de la base Yahoo Finance : fonction `pdfetch::pdfetch_YAHOO()`.

Vous avez pour mission de modéliser le risque présent sur le marché du CAC40.

Question 0

Importer le setup de l'analyse.

Question 1

Importer les données du CAC40 à partir de la base Yahoo Finance.

Question 2

Les données de cours boursier du CAC40 sont-elles stationnaires ?

Question 3

Stationnariser la série si besoin

Question 4

Modéliser le risque à partir d'une mesure de Value at Risk non-conditionnelle historique, de Student (normale pour une version plus simple) en statique et en fenêtre glissante de 252 jours. Commentez.

Question 5

Trouver le meilleur modèle (S)ARIMA pour modéliser l'espérance conditionnelle de la série stationnaire à partir du critère d'information BIC

Question 6

Les résidus de ce modèle sont-ils normaux ? Sont-ils auto-corrélés ? Les résidus au carré sont-ils auto-corrélés ? Y a-t-il des effets ARCH dans les résidus. Qu'en concluez-vous pour la suite de l'analyse.

Question 7

Estimer la volatilité conditionnelle à partir de 3 modèles (minimum) différents. Expliquez votre processus de sélection et de validation des modèles utilisés.

Question 8

Calculer la VaR associée à ces 3 modèles.

Question 9

Effectuer les tests de Backtests sur tous les modèles de VaR estimés. Quelles sont les conclusions ?

Question 10

Parmi tous les modèles de VaR validés (ou parmi tous les modèles s'il n'y a pas assez de modèles validés) quel est le modèle que vous choisissez pour modéliser le risque du CAC40 ?

Exercice 2 : Prédire le FTSE100

Votre patron vous demande de prédire les futurs rendements du FTSE100 ainsi que les risques de perte maximum pour un niveau de confiance de 99%. Expliquer votre démarche et abalyser les résultats.

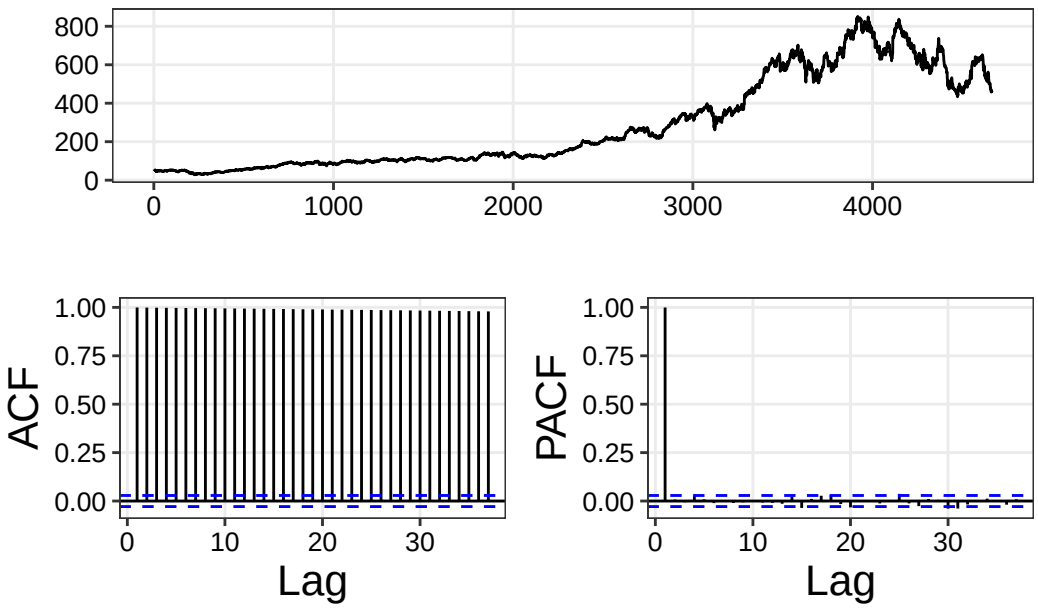
Exercice 3 : LVMH

Votre collège part à la retraite et vous transmet le projet sur lequel il travaillait : modéliser le risque des prix de l'action LVMH. Commentez les résultats et les méthodes qu'il utilise.

```
rm(list = ls())
source(here::here("02-codes", "utils", "setup.R"))
```

```
lvmh <-
  pdfetch::pdfetch_YAHOO(
    "MC.PA",
    fields = "adjclose",
    from = as.Date("2008-01-03"),
    to = as.Date("2026-03-27")
  )
```

```
forecast::ggtsdisplay(lvmh, point = FALSE)
```




```
plot(lvmh)
```

```r
urca::ur.df(lvmh, "trend", lags = 20, selectlags = "AIC") |> summary()
```

```
###############################################
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression trend


Call:
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)

Residuals:
    Min      1Q  Median      3Q     Max
-46.415  -1.581  -0.043   1.628  83.432

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.0069552  0.2142287  -0.032  0.97410
z.lag.1      -0.0016869  0.0009619  -1.754  0.07955 .
tt            0.0002516  0.0001712   1.470  0.14173
z.diff.lag1  -0.0154563  0.0146999  -1.051  0.29310
```

```
z.diff.lag2   0.0103340  0.0146878   0.704  0.48173
z.diff.lag3  -0.0393151  0.0146881  -2.677  0.00746 **
z.diff.lag4  -0.0074513  0.0146900  -0.507  0.61201
z.diff.lag5   0.0134216  0.0146851   0.914  0.36079
z.diff.lag6  -0.0062563  0.0146876  -0.426  0.67016
z.diff.lag7   0.0103925  0.0146877   0.708  0.47925
z.diff.lag8   0.0043109  0.0146892   0.293  0.76918
z.diff.lag9   0.0074862  0.0146896   0.510  0.61034
z.diff.lag10 -0.0079605  0.0147054  -0.541  0.58831
z.diff.lag11  0.0106393  0.0147066   0.723  0.46945
z.diff.lag12  0.0121689  0.0147075   0.827  0.40806
z.diff.lag13 -0.0250172  0.0147090  -1.701  0.08905 .
z.diff.lag14  0.0423766  0.0147134   2.880  0.00399 **
z.diff.lag15 -0.0126622  0.0147144  -0.861  0.38954
z.diff.lag16 -0.0415920  0.0147152  -2.826  0.00473 **
z.diff.lag17 -0.0400168  0.0147267  -2.717  0.00661 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.622 on 4624 degrees of freedom
Multiple R-squared:  0.00938,   Adjusted R-squared:  0.005309
F-statistic: 2.304 on 19 and 4624 DF,  p-value: 0.001053


Value of test-statistic is: -1.7537 1.388 1.5773

Critical values for test statistics:
      1pct  5pct 10pct
tau3 -3.96 -3.41 -3.12
phi2  6.09  4.68  4.03
phi3  8.27  6.25  5.34
```

```r
urca::ur.df(lvmh, "drift", lags = 20, selectlags = "AIC") |> summary()
```

```
###############################################
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression drift


Call:
```

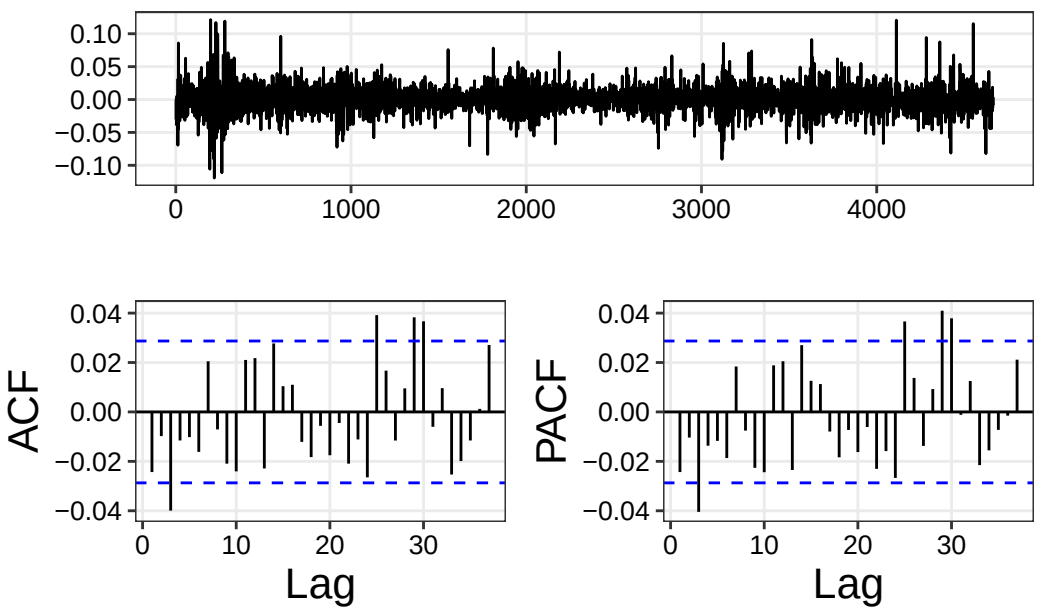
```
lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)

Residuals:
    Min      1Q  Median      3Q     Max 
-46.265  -1.577  -0.065   1.659  83.381 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)    
(Intercept)   0.2145522  0.1522562   1.409  0.15886    
z.lag.1      -0.0004062  0.0004073  -0.997  0.31869    
z.diff.lag1  -0.0162387  0.0146921  -1.105  0.26910    
z.diff.lag2   0.0095838  0.0146808   0.653  0.51391    
z.diff.lag3  -0.0400533  0.0146813  -2.728  0.00639 ** 
z.diff.lag4  -0.0082080  0.0146828  -0.559  0.57617    
z.diff.lag5   0.0126890  0.0146784   0.864  0.38738    
z.diff.lag6  -0.0069928  0.0146809  -0.476  0.63387    
z.diff.lag7   0.0096636  0.0146812   0.658  0.51042    
z.diff.lag8   0.0035924  0.0146829   0.245  0.80673    
z.diff.lag9   0.0067600  0.0146831   0.460  0.64526    
z.diff.lag10 -0.0086569  0.0146996  -0.589  0.55594    
z.diff.lag11  0.0099529  0.0147010   0.677  0.49843    
z.diff.lag12  0.0114887  0.0147021   0.781  0.43459    
z.diff.lag13 -0.0257163  0.0147032  -1.749  0.08035 .  
z.diff.lag14  0.0417036  0.0147081   2.835  0.00460 ** 
z.diff.lag15 -0.0133223  0.0147094  -0.906  0.36514    
z.diff.lag16 -0.0422468  0.0147103  -2.872  0.00410 ** 
z.diff.lag17 -0.0406477  0.0147223  -2.761  0.00579 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.623 on 4625 degrees of freedom
Multiple R-squared:  0.008917,  Adjusted R-squared:  0.00506 
F-statistic: 2.312 on 18 and 4625 DF,  p-value: 0.001299


Value of test-statistic is: -0.9973 1.0019

Critical values for test statistics:
      1pct  5pct 10pct
tau2 -3.43 -2.86 -2.57
phi1  6.43  4.59  3.78
```

```r
urca::ur.df(lvmh, "none", lags = 20, selectlags = "AIC") |> summary()
```

```
###############################################
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression none


Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)

Residuals:
    Min      1Q  Median      3Q     Max 
-46.310  -1.453   0.080   1.786  83.306 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)   
z.lag.1       0.0000351  0.0002605   0.135  0.89284   
z.diff.lag1  -0.0162523  0.0146936  -1.106  0.26875   
z.diff.lag2   0.0095757  0.0146824   0.652  0.51431   
z.diff.lag3  -0.0400643  0.0146829  -2.729  0.00638 **
z.diff.lag4  -0.0082022  0.0146844  -0.559  0.57648   
z.diff.lag5   0.0126976  0.0146800   0.865  0.38710   
z.diff.lag6  -0.0069851  0.0146824  -0.476  0.63428   
z.diff.lag7   0.0096729  0.0146827   0.659  0.51006   
z.diff.lag8   0.0035989  0.0146845   0.245  0.80640   
z.diff.lag9   0.0067619  0.0146847   0.460  0.64520   
z.diff.lag10 -0.0086566  0.0147012  -0.589  0.55600   
z.diff.lag11  0.0099569  0.0147026   0.677  0.49830   
z.diff.lag12  0.0114885  0.0147036   0.781  0.43465   
z.diff.lag13 -0.0257235  0.0147047  -1.749  0.08030 . 
z.diff.lag14  0.0417076  0.0147096   2.835  0.00460 **
z.diff.lag15 -0.0133382  0.0147110  -0.907  0.36462   
z.diff.lag16 -0.0422574  0.0147118  -2.872  0.00409 **
z.diff.lag17 -0.0406430  0.0147239  -2.760  0.00580 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.623 on 4626 degrees of freedom
Multiple R-squared:  0.008667,	Adjusted R-squared:  0.00481 
F-statistic: 2.247 on 18 and 4626 DF,  p-value: 0.001876
```

```
Value of test-statistic is: 0.1347

Critical values for test statistics:
      1pct  5pct 10pct
tau1 -2.58 -1.95 -1.62
```

```r
rend_lvmh <-
  log(lvmh) - log(lag(lvmh)) |>
  na.omit()

forecast::ggtsdisplay(rend_lvmh, point = FALSE)
```




```r
moments::jarque.test(as.numeric(rend_lvmh))
```

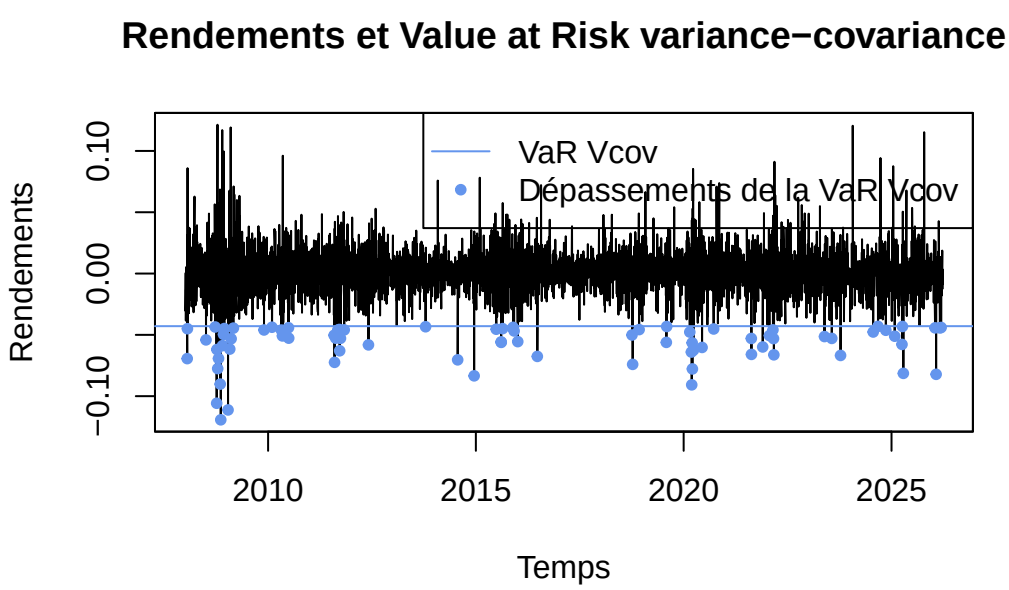
```
    Jarque-Bera Normality Test

data:  as.numeric(rend_lvmh)
JB = 4287, p-value < 0.00000000000000022
alternative hypothesis: greater
```

```r
t_couv <- 0.01
```

```r
VaR_VCov <- (-(mean(rend_lvmh)+sd(rend_lvmh)*qnorm(t_couv))) |> print()
```

```
[1] 0.04291616
```

```r
plot(as.zoo(rend_lvmh),
      main = "Rendements et Value at Risk variance-covariance",
      xlab = "Temps",
      ylab = "Rendements")
  abline(h= -VaR_VCov, col="cornflowerblue")
  points(as.zoo(rend_lvmh[as.numeric(rend_lvmh) < -VaR_VCov]), pch=20,
  col="cornflowerblue")
  legend("topright", legend = c("VaR Vcov","Dépassements de la VaR Vcov"),
         pch = c(NA,20), lty = c(1, NA), col = "cornflowerblue")
```

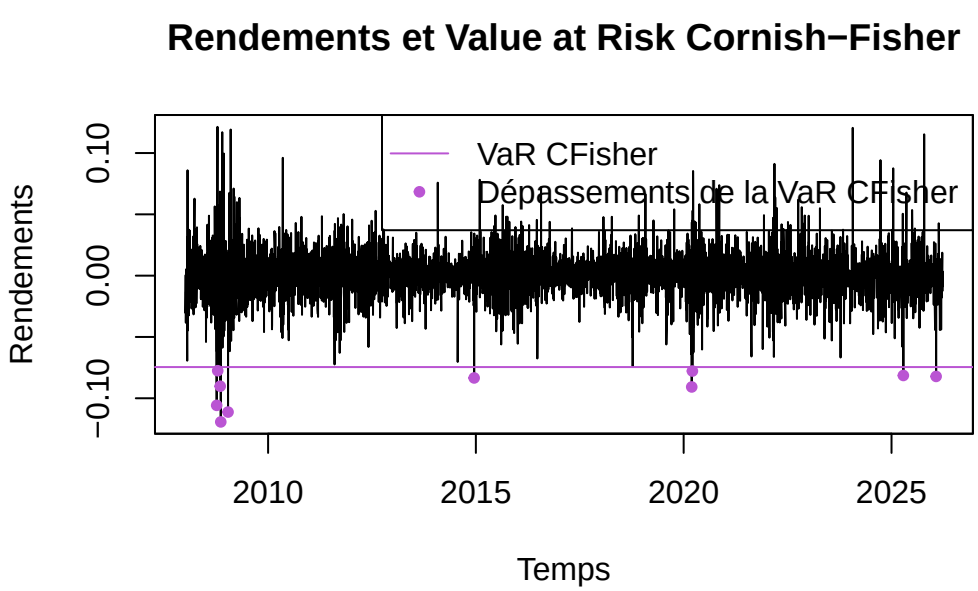



```r
K <- kurtosis(rend_lvmh)
S <- skewness(rend_lvmh)

k <-
as.numeric(qnorm(t_couv)+(qnorm(t_couv)^2-1)*(S/6)+(qnorm(t_couv)^3-3*qnorm(t_couv))*(K/24)+
```

```r
VaR_CFisher<- as.numeric(-(mean(rend_lvmh)+sd(rend_lvmh)*k)) |> print()
```

```
[1] 0.07455505
```

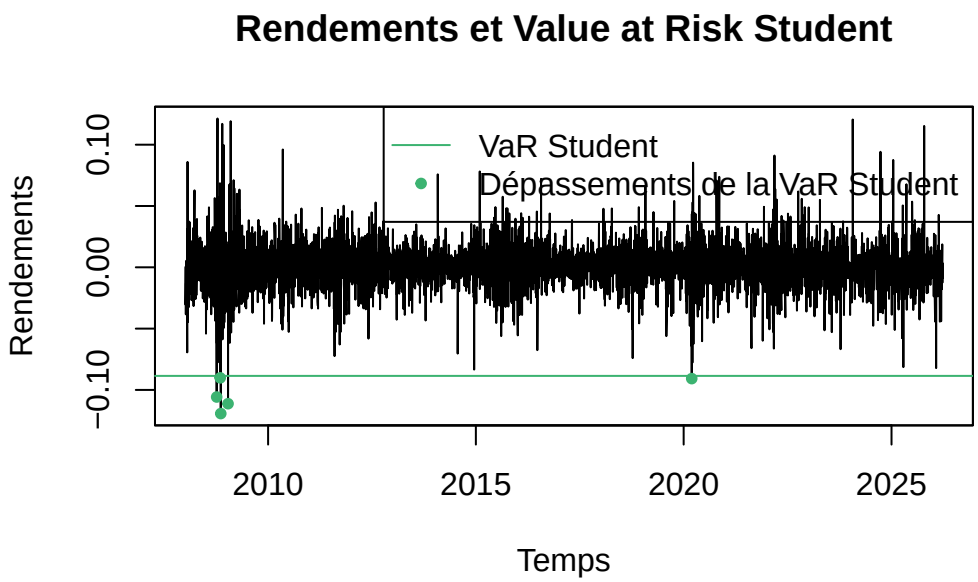
```r
plot(as.zoo(rend_lvmh),
      main = "Rendements et Value at Risk Cornish-Fisher",
      xlab = "Temps",
      ylab = "Rendements")
  abline(h= -VaR_CFisher, col="mediumorchid")
  points(as.zoo(rend_lvmh[as.numeric(rend_lvmh) < -VaR_CFisher]), pch=20,
  col="mediumorchid")
  legend("topright", legend = c("VaR CFisher","Dépassements de la VaR
  CFisher"),
         pch = c(NA,20), lty = c(1, NA), col = "mediumorchid")
```




```r
param.sd <- fitdist(distribution = 'std', x = rend_lvmh)$pars
quantile.sd <- qt(t_couv, df=param.sd["shape"])

VaR_student <-
(-mean(rend_lvmh)-sd(rend_lvmh)*quantile.sd*sqrt((quantile.sd-2)/quantile.sd))
|> print()
```

```
[1] 0.08855781
```

```r
plot(as.zoo(rend_lvmh),
      main = "Rendements et Value at Risk Student",
      xlab = "Temps",
      ylab = "Rendements")
  abline(h= -VaR_student, col="mediumseagreen")
  points(as.zoo(rend_lvmh[as.numeric(rend_lvmh) < -VaR_student]), pch=20,
  col="mediumseagreen")
  legend("topright", legend = c("VaR Student","Dépassements de la VaR
  Student"),
         pch = c(NA,20), lty = c(1, NA), col = "mediumseagreen")
```




```r
n_backtest <- 2000

taille_sample_estimation <- length(rend_lvmh) - n_backtest

sample_estimation <- rend_lvmh[1:taille_sample_estimation]

sample_backtest <- rend_lvmh[(taille_sample_estimation+1):length(rend_lvmh)]
```

```r
VaR_VCov_estimation <-
(-(mean(sample_estimation)+sd(sample_estimation)*qnorm(t_couv)))

test_VaR_VCov <-
  rugarch::VaRTest(
    alpha = t_couv,
    actual = sample_backtest,
    VaR = rep(-VaR_VCov_estimation, times = length(sample_backtest)),
    conf.level = 0.95
  )

tibble(
  name = names(unlist(test_VaR_VCov)),
  value = unlist(test_VaR_VCov)
) |>
  print(n = 20)
```

```
# A tibble: 12 x 2
   name            value
   <chr>           <chr>
 1 expected.exceed 20
 2 actual.exceed   34
 3 uc.H0           Correct Exceedances
 4 uc.LRstat       8.18194510909123
 5 uc.critical     3.84145882069412
 6 uc.LRp          0.00423093684953424
 7 uc.Decision     Reject H0
 8 cc.H0           Correct Exceedances & Independent
 9 cc.LRstat       10.4255909694824
10 cc.critical     5.99146454710798
11 cc.LRp          0.00544642698307951
12 cc.Decision     Reject H0
```

```r
HIT_VCov <-
  if_else(sample_backtest < -VaR_VCov_estimation, 1, 0)

HIT_centre_VCov <- HIT_VCov - t_couv

stats::Box.test(HIT_centre_VCov, lag = 5, type = "Ljung-Box")
```

```
    Box-Ljung test
```

```
data:  HIT_centre_VCov
X-squared = 28.608, df = 5, p-value = 0.00002767
```

```r
df <- tibble(
  HIT_centre = HIT_centre_VCov,
  rend = sample_backtest |> as.numeric(),
  rend2 = sample_backtest**2 |> as.numeric()
)

ARDL::ardl(
  HIT_centre ~ rend + rend2,
  data = df,
  order = 5
) |>
  summary()
```

```
Time series regression with "ts" data:
Start = 6, End = 2000

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
     Min       1Q   Median       3Q      Max
-0.71827 -0.02552  0.00925  0.02990  0.73041

Coefficients:
                  Estimate Std. Error t value             Pr(>|t|)
(Intercept)      -0.018914   0.003056  -6.190       0.000000000731 ***
L(HIT_centre, 1)  0.003505   0.022494   0.156               0.8762
L(HIT_centre, 2) -0.016734   0.022497  -0.744               0.4571
L(HIT_centre, 3)  0.017929   0.022536   0.796               0.4264
L(HIT_centre, 4) -0.004361   0.022506  -0.194               0.8464
L(HIT_centre, 5)  0.004296   0.022482   0.191               0.8485
rend             -3.152929   0.120388 -26.190 < 0.0000000000000002 ***
L(rend, 1)       -0.185045   0.139890  -1.323               0.1861
L(rend, 2)       -0.317955   0.139848  -2.274               0.0231 *
L(rend, 3)        0.058905   0.140008   0.421               0.6740
L(rend, 4)        0.021289   0.140771   0.151               0.8798
L(rend, 5)       -0.024329   0.140435  -0.173               0.8625
```

```
rend2            76.422467   2.662758  28.700 < 0.0000000000000002 ***
L(rend2, 1)       0.211886   3.165889   0.067               0.9466
L(rend2, 2)       6.670499   3.144203   2.122               0.0340 *
L(rend2, 3)       0.909282   3.139156   0.290               0.7721
L(rend2, 4)      -2.119300   3.147434  -0.673               0.5008
L(rend2, 5)      -3.543887   3.145991  -1.126               0.2601
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09867 on 1977 degrees of freedom
Multiple R-squared:  0.4241,    Adjusted R-squared:  0.4192
F-statistic: 85.64 on 17 and 1977 DF,  p-value: < 0.00000000000000022
```

```r
K <- kurtosis(sample_estimation)
S <- skewness(sample_estimation)

k <-
as.numeric(qnorm(t_couv)+(qnorm(t_couv)^2-1)*(S/6)+(qnorm(t_couv)^3-3*qnorm(t_couv))*(K/24)+

VaR_CFisher_estimation <-
as.numeric(-(mean(sample_estimation)+sd(sample_estimation)*k))

test_VaR_CFisher <-
  rugarch::VaRTest(
    alpha = t_couv,
    actual = sample_backtest,
    VaR = rep(-VaR_CFisher_estimation, times = length(sample_backtest)),
    conf.level = 0.95
  )

tibble(
  name = names(unlist(test_VaR_CFisher)),
  value = unlist(test_VaR_CFisher)
) |>
  print(n = 20)
```

```
# A tibble: 12 x 2
   name            value
   <chr>           <chr>
 1 expected.exceed 20
 2 actual.exceed   4
 3 uc.H0           Correct Exceedances
 4 uc.LRstat       19.2534427663781
```

```
 5 uc.critical     3.84145882069412
 6 uc.LRp          0.0000114463780930274
 7 uc.Decision     Reject H0
 8 cc.H0           Correct Exceedances & Independent
 9 cc.LRstat       19.2694828773756
10 cc.critical     5.99146454710798
11 cc.LRp          0.0000654161466900893
12 cc.Decision     Reject H0
```

```r
HIT_CFisher <-
  if_else(sample_backtest < -VaR_CFisher_estimation, 1, 0)

HIT_centre_CFisher <- HIT_VCov - t_couv

stats::Box.test(HIT_centre_VCov, lag = 5, type = "Ljung-Box")
```

```
    Box-Ljung test

data:  HIT_centre_VCov
X-squared = 28.608, df = 5, p-value = 0.00002767
```

```r
df <- tibble(
  HIT_centre = HIT_centre_CFisher,
  rend = sample_backtest |> as.numeric(),
  rend2 = sample_backtest**2 |> as.numeric()
)

ARDL::ardl(
  HIT_centre ~ rend + rend2,
  data = df,
  order = 5
) |>
  summary()
```

```
Time series regression with "ts" data:
Start = 6, End = 2000

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)
```

```
Residuals:
     Min       1Q   Median       3Q      Max 
-0.71827 -0.02552  0.00925  0.02990  0.73041 

Coefficients:
                  Estimate Std. Error t value             Pr(>|t|)    
(Intercept)      -0.018914   0.003056  -6.190       0.000000000731 ***
L(HIT_centre, 1)  0.003505   0.022494   0.156               0.8762    
L(HIT_centre, 2) -0.016734   0.022497  -0.744               0.4571    
L(HIT_centre, 3)  0.017929   0.022536   0.796               0.4264    
L(HIT_centre, 4) -0.004361   0.022506  -0.194               0.8464    
L(HIT_centre, 5)  0.004296   0.022482   0.191               0.8485    
rend             -3.152929   0.120388 -26.190 < 0.0000000000000002 ***
L(rend, 1)       -0.185045   0.139890  -1.323               0.1861    
L(rend, 2)       -0.317955   0.139848  -2.274               0.0231 *  
L(rend, 3)        0.058905   0.140008   0.421               0.6740    
L(rend, 4)        0.021289   0.140771   0.151               0.8798    
L(rend, 5)       -0.024329   0.140435  -0.173               0.8625    
rend2            76.422467   2.662758  28.700 < 0.0000000000000002 ***
L(rend2, 1)       0.211886   3.165889   0.067               0.9466    
L(rend2, 2)       6.670499   3.144203   2.122               0.0340 *  
L(rend2, 3)       0.909282   3.139156   0.290               0.7721    
L(rend2, 4)      -2.119300   3.147434  -0.673               0.5008    
L(rend2, 5)      -3.543887   3.145991  -1.126               0.2601    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09867 on 1977 degrees of freedom
Multiple R-squared:  0.4241,	Adjusted R-squared:  0.4192 
F-statistic: 85.64 on 17 and 1977 DF,  p-value: < 0.00000000000000022
```

```r
param.sd <- fitdist(distribution = 'std', x = sample_estimation)$pars
quantile.sd <- qt(t_couv, df=param.sd["shape"])

VaR_student_estimation <-
(-mean(sample_estimation)-sd(sample_estimation)*quantile.sd*sqrt((quantile.sd-2)/quantile.sd

test_VaR_student <-
  rugarch::VaRTest(
    alpha = t_couv,
    actual = sample_backtest,
    VaR = rep(-VaR_student_estimation, times = length(sample_backtest)),
    conf.level = 0.95
```

```r
  )

tibble(
  name = names(unlist(test_VaR_student)),
  value = unlist(test_VaR_student)
) |>
  print(n = 20)
```

```
# A tibble: 12 x 2
   name            value
   <chr>           <chr>
 1 expected.exceed 20
 2 actual.exceed   1
 3 uc.H0           Correct Exceedances
 4 uc.LRstat       32.1902782785452
 5 uc.critical     3.84145882069412
 6 uc.LRp          0.00000001397884619169 79
 7 uc.Decision     Reject H0
 8 cc.H0           Correct Exceedances & Independent
 9 cc.LRstat       32.1912792795877
10 cc.critical     5.99146454710798
11 cc.LRp          0.00000010227100455395 5
12 cc.Decision     Reject H0
```

```r
HIT_student <-
  if_else(sample_backtest < -VaR_student_estimation, 1, 0)

HIT_centre_student <- HIT_VCov - t_couv

stats::Box.test(HIT_centre_student, lag = 5, type = "Ljung-Box")
```

```
    Box-Ljung test

data:  HIT_centre_student
X-squared = 28.608, df = 5, p-value = 0.00002767
```

```r
df <- tibble(
  HIT_centre = HIT_centre_student,
  rend = sample_backtest |> as.numeric(),
  rend2 = sample_backtest**2 |> as.numeric()
)
```

```r
ARDL::ardl(
  HIT_centre ~ rend + rend2,
  data = df,
  order = 5
) |>
  summary()
```

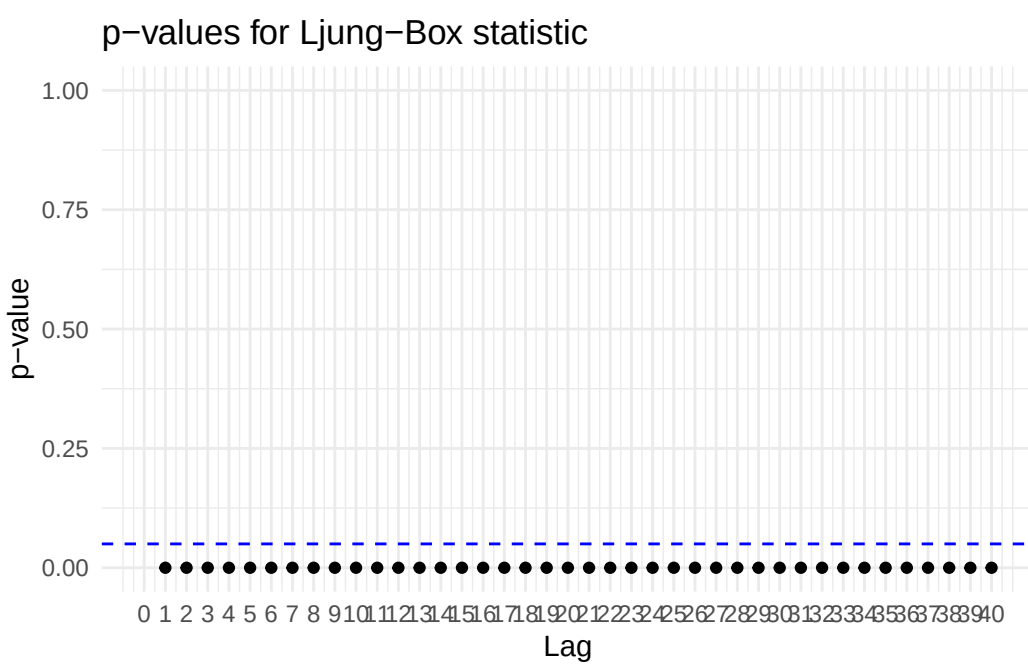
```
Time series regression with "ts" data:
Start = 6, End = 2000

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
     Min       1Q   Median       3Q      Max
-0.71827 -0.02552  0.00925  0.02990  0.73041

Coefficients:
                  Estimate Std. Error t value             Pr(>|t|)
(Intercept)      -0.018914   0.003056  -6.190       0.000000000731 ***
L(HIT_centre, 1)  0.003505   0.022494   0.156               0.8762
L(HIT_centre, 2) -0.016734   0.022497  -0.744               0.4571
L(HIT_centre, 3)  0.017929   0.022536   0.796               0.4264
L(HIT_centre, 4) -0.004361   0.022506  -0.194               0.8464
L(HIT_centre, 5)  0.004296   0.022482   0.191               0.8485
rend             -3.152929   0.120388 -26.190 < 0.0000000000000002 ***
L(rend, 1)       -0.185045   0.139890  -1.323               0.1861
L(rend, 2)       -0.317955   0.139848  -2.274               0.0231 *
L(rend, 3)        0.058905   0.140008   0.421               0.6740
L(rend, 4)        0.021289   0.140771   0.151               0.8798
L(rend, 5)       -0.024329   0.140435  -0.173               0.8625
rend2            76.422467   2.662758  28.700 < 0.0000000000000002 ***
L(rend2, 1)       0.211886   3.165889   0.067               0.9466
L(rend2, 2)       6.670499   3.144203   2.122               0.0340 *
L(rend2, 3)       0.909282   3.139156   0.290               0.7721
L(rend2, 4)      -2.119300   3.147434  -0.673               0.5008
L(rend2, 5)      -3.543887   3.145991  -1.126               0.2601
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.09867 on 1977 degrees of freedom
Multiple R-squared:  0.4241,    Adjusted R-squared:  0.4192
F-statistic: 85.64 on 17 and 1977 DF,  p-value: < 0.00000000000000022
```

```r
best_arma <- forecast::auto.arima(rend_lvmh)

ar <- best_arma$arma[1]
ma <- ar <- best_arma$arma[2]

best_arma$arma[c(1,6,2)] |>  print()
```

```
[1] 1 0 2
```

```r
LSTS::Box.Ljung.Test(best_arma$residuals**2, lag = 40)
```




```r
fDMA::archtest(as.numeric(best_arma$residuals**2), lag = 20)
```

```
    Engle's LM ARCH Test

data:  as.numeric(best_arma$residuals^2)
statistic = 213.69, lag = 20, p-value < 0.00000000000000022
alternative hypothesis: ARCH effects of order 20 are present
```

```r
garch11_spec <-
  rugarch::ugarchspec(
    variance.model =
      list(
        model = "sGARCH",
        garchOrder = c(1,1),
        variance.targeting = TRUE
      ),
    mean.model =
      list(
        armaOrder = c(ar,ma),
        include.mean = TRUE
      ),
    distribution.model = "sged"
  )

garch11_fit <-
  rugarch::ugarchfit(
    garch11_spec,
    rend_lvmh
  )

garch11_fit
```

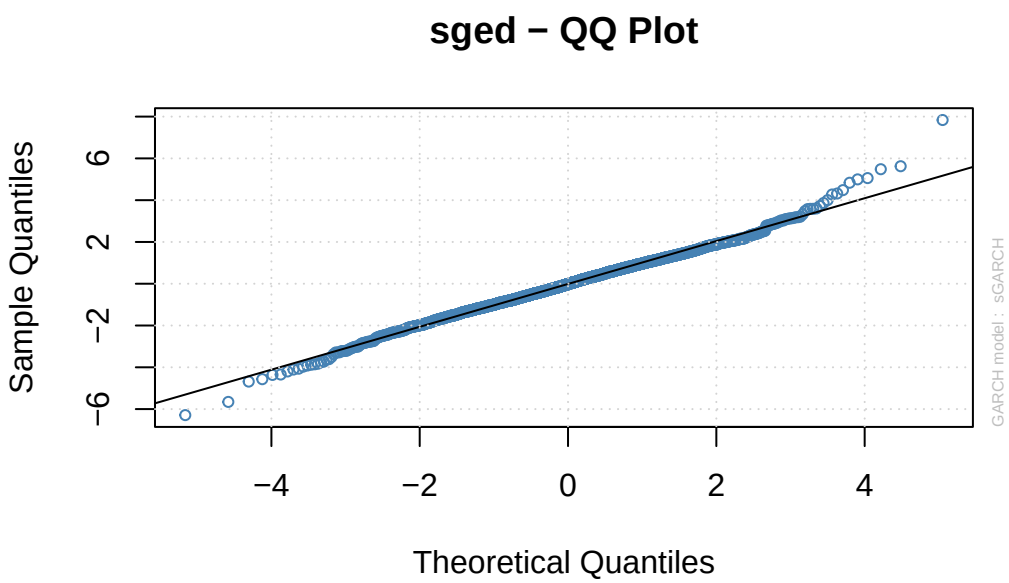
```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model : sGARCH(1,1)
Mean Model  : ARFIMA(2,0,2)
Distribution    : sged

Optimal Parameters
------------------------------------
        Estimate  Std. Error   t value Pr(>|t|)
mu      0.000703    0.000224    3.1406 0.001686
ar1     0.064753    0.010373    6.2422 0.000000
ar2     0.413149    0.107128    3.8566 0.000115
ma1    -0.102676    0.015025   -6.8338 0.000000
ma2    -0.432068    0.104995   -4.1151 0.000039
```

```
alpha1  0.052968    0.002201   24.0613 0.000000
beta1   0.930339    0.000724 1284.2748 0.000000
skew    0.986787    0.019404   50.8544 0.000000
shape   1.227260    0.031118   39.4391 0.000000
omega   0.000006          NA        NA       NA

Robust Standard Errors:
        Estimate  Std. Error   t value Pr(>|t|)
mu      0.000703    0.000254    2.7677 0.005645
ar1     0.064753    0.020210    3.2040 0.001356
ar2     0.413149    0.056703    7.2862 0.000000
ma1    -0.102676    0.021368   -4.8051 0.000002
ma2    -0.432068    0.056878   -7.5963 0.000000
alpha1  0.052968    0.002267   23.3613 0.000000
beta1   0.930339    0.000610 1525.8507 0.000000
skew    0.986787    0.024230   40.7253 0.000000
shape   1.227260    0.038801   31.6299 0.000000
omega   0.000006          NA        NA       NA

LogLikelihood : 12447.91

Information Criteria
------------------------------------

Akaike       -5.3340
Bayes        -5.3216
Shibata      -5.3340
Hannan-Quinn -5.3296

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic p-value
Lag[1]                     0.6109  0.4344
Lag[2*(p+q)+(p+q)-1][11]   5.3350  0.8670
Lag[4*(p+q)+(p+q)-1][19]   8.4358  0.7301
d.o.f=4
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                        statistic p-value
Lag[1]                 0.00002504  0.9960
Lag[2*(p+q)+(p+q)-1][5] 0.34997915  0.9782
```

```
Lag[4*(p+q)+(p+q)-1][9] 1.18065656  0.9775
d.o.f=2

Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[3]   0.09666 0.500 2.000  0.7559
ARCH Lag[5]   0.61768 1.440 1.667  0.8484
ARCH Lag[7]   0.93422 2.315 1.543  0.9240

Nyblom stability test
------------------------------------
Joint Statistic:  2.977
Individual Statistics:
mu     0.2527
ar1    0.1285
ar2    0.1806
ma1    0.1460
ma2    0.1965
alpha1 0.4902
beta1  0.6451
skew   0.1343
shape  1.3999

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:         2.1 2.32 2.82
Individual Statistic:    0.35 0.47 0.75

Sign Bias Test
------------------------------------
                   t-value     prob sig
Sign Bias            1.579 0.114486
Negative Sign Bias   1.599 0.109883
Positive Sign Bias   2.699 0.006981 ***
Joint Effect        10.505 0.014726  **


Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     33.19      0.02287
2    30     33.08      0.27436
3    40     47.75      0.15883
```

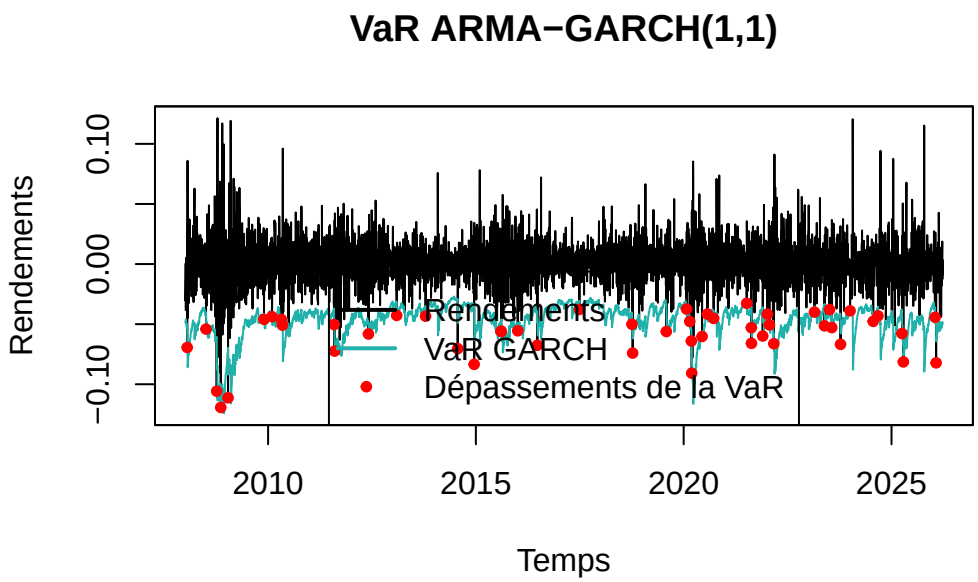
```
4     50      60.40       0.12738

Elapsed time : 1.115992
```

```r
plot(garch11_fit, which = 9)
```




```r
# Calculer la VaR à partir du modèle EGARCH
VaR_GARCH <- -(as.numeric(quantile(garch11_fit, probs = t_couv)))

# Représenter les rendements et la VaR
plot(
  # Combiner dans un seul df les rendements et la VaR
  as.zoo(cbind(rend_lvmh,-VaR_GARCH)),
  screens=1,
  col = c("black","lightseagreen"),
  main="VaR ARMA-GARCH(1,1)",
  xlab = "Temps",
  ylab = "Rendements"
)
points(as.zoo(rend_lvmh[as.numeric(rend_lvmh) < -VaR_GARCH]), pch=20,
col="red")
```

```
legend("bottom",c("Rendements","VaR GARCH", "Dépassements de la VaR"),
         col = c("black","lightseagreen", 'red'),lty = c(1,1,NA),lwd =
         c(2,2,NA),pch = c(NA,NA,20))
```




```
garch11_estimation_fit <-
  rugarch::ugarchfit(
    garch11_spec,
    rend_lvmh,
    out.sample = n_backtest
  )

forecast_garch11 <-
  rugarch::ugarchforecast(
    garch11_estimation_fit,
    data = rend_lvmh,
    n.roll = n_backtest-1,
    n.ahead = 1
  )

VaR_GARCH_backtest <- -(as.numeric(quantile(forecast_garch11, probs = 0.01)))

test_VaR_GARCH<-
```

```r
  rugarch::VaRTest(
    alpha = t_couv,
    actual = sample_backtest,
    VaR = -VaR_GARCH_backtest,
    conf.level = 0.95
  )

tibble(
  name = names(unlist(test_VaR_GARCH)),
  value = unlist(test_VaR_GARCH)
) |>
  print(n = 20)
```

```
# A tibble: 12 x 2
   name            value
   <chr>           <chr>
 1 expected.exceed 20
 2 actual.exceed   30
 3 uc.H0           Correct Exceedances
 4 uc.LRstat       4.37849677769573
 5 uc.critical     3.84145882069412
 6 uc.LRp          0.0363950816720029
 7 uc.Decision     Reject H0
 8 cc.H0           Correct Exceedances & Independent
 9 cc.LRstat       7.41028267304434
10 cc.critical     5.99146454710798
11 cc.LRp          0.0245967407041665
12 cc.Decision     Reject H0
```

```r
HIT <- if_else(sample_backtest < -VaR_GARCH_backtest, 1, 0)

HIT_centre <- HIT - 0.01

stats::Box.test(HIT_centre, lag = 5, type = "Ljung-Box")
```

```
    Box-Ljung test

data:  HIT_centre
X-squared = 7.8284, df = 5, p-value = 0.166
```

```r
df <-
  tibble(
    HIT_centre = HIT_centre,
    rend = sample_backtest,
    rend2 = sample_backtest**2
  )

ARDL::ardl(
  HIT_centre ~ rend + rend2,
  data = df,
  order = 5
) |>
  summary()
```

```
Time series regression with "ts" data:
Start = 6, End = 2000

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
     Min       1Q   Median       3Q      Max
-0.58429 -0.02453  0.00551  0.02254  0.84485

Coefficients:
                  Estimate Std. Error t value             Pr(>|t|)
(Intercept)      -0.008767   0.002990  -2.932              0.00341 **
L(HIT_centre, 1)  0.009453   0.022519   0.420              0.67470
L(HIT_centre, 2) -0.028049   0.022499  -1.247              0.21267
L(HIT_centre, 3)  0.004019   0.022500   0.179              0.85826
L(HIT_centre, 4) -0.003649   0.022482  -0.162              0.87109
L(HIT_centre, 5)  0.006732   0.022462   0.300              0.76444
rend             -2.644755   0.123224 -21.463 < 0.0000000000000002 ***
L(rend, 1)       -0.187271   0.137030  -1.367              0.17189
L(rend, 2)       -0.202942   0.136924  -1.482              0.13846
L(rend, 3)       -0.176489   0.136651  -1.292              0.19667
L(rend, 4)        0.011667   0.137550   0.085              0.93241
L(rend, 5)        0.063206   0.137068   0.461              0.64476
rend2            62.710643   2.715539  23.093 < 0.0000000000000002 ***
L(rend2, 1)       0.657909   3.056801   0.215              0.82961
L(rend2, 2)      -5.265765   3.034858  -1.735              0.08288 .
```

```
L(rend2, 3)      -4.618037   3.029826  -1.524              0.12762
L(rend2, 4)      -5.419415   3.037485  -1.784              0.07455 .
L(rend2, 5)      -5.051534   3.041040  -1.661              0.09685 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1007 on 1977 degrees of freedom
Multiple R-squared:  0.3218,    Adjusted R-squared:  0.3159
F-statistic: 55.17 on 17 and 1977 DF,  p-value: < 0.00000000000000022
```

```r
egarch11_spec <-
  rugarch::ugarchspec(
    variance.model =
      list(
        model = "eGARCH",
        garchOrder = c(1,1),
        variance.targeting = TRUE
      ),
    mean.model =
      list(
        armaOrder = c(ar,ma),
        include.mean = TRUE
      ),
    distribution.model = "sged"
  )

egarch11_fit <-
  rugarch::ugarchfit(
    egarch11_spec,
    rend_lvmh
  )

egarch11_fit
```

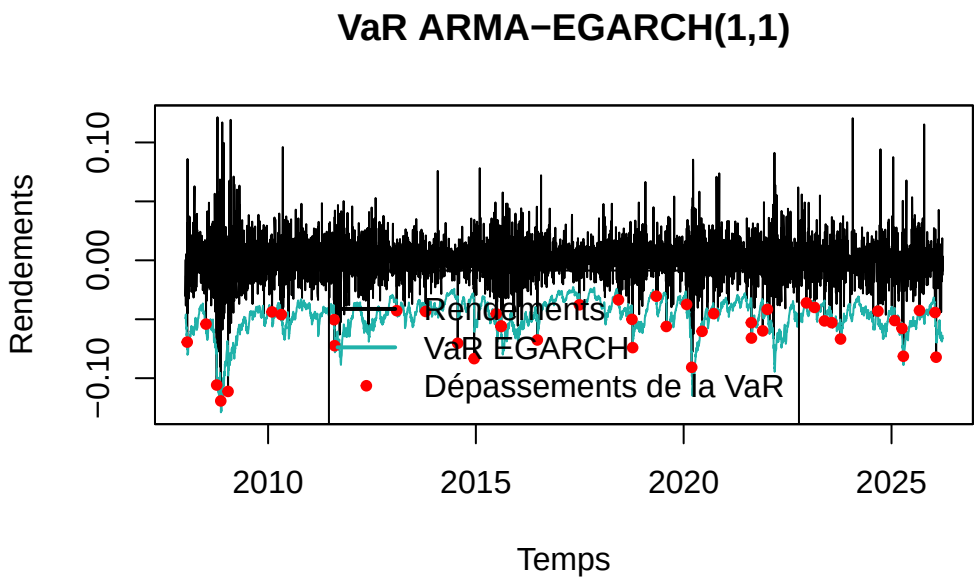
```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model : eGARCH(1,1)
Mean Model  : ARFIMA(2,0,2)
```

```
Distribution    : sged

Optimal Parameters
------------------------------------
        Estimate  Std. Error    t value Pr(>|t|)
mu      0.000168    0.000151     1.1115  0.26636
ar1     1.157150    0.038920    29.7312  0.00000
ar2    -0.640724    0.016794   -38.1512  0.00000
ma1    -1.173413    0.040785   -28.7710  0.00000
ma2     0.630219    0.015902    39.6322  0.00000
alpha1 -0.082114    0.006068   -13.5315  0.00000
beta1   0.983916    0.000025 39265.0772  0.00000
gamma1  0.083800    0.007716    10.8602  0.00000
skew    0.972248    0.016505    58.9047  0.00000
shape   1.277337    0.034561    36.9586  0.00000
omega  -0.128127          NA         NA       NA

Robust Standard Errors:
        Estimate  Std. Error    t value Pr(>|t|)
mu      0.000168    0.000153     1.0998  0.27142
ar1     1.157150    0.012417    93.1918  0.00000
ar2    -0.640724    0.005363  -119.4654  0.00000
ma1    -1.173413    0.013215   -88.7925  0.00000
ma2     0.630219    0.005817   108.3433  0.00000
alpha1 -0.082114    0.007372   -11.1388  0.00000
beta1   0.983916    0.000025 39391.9624  0.00000
gamma1  0.083800    0.009233     9.0765  0.00000
skew    0.972248    0.017892    54.3407  0.00000
shape   1.277337    0.042645    29.9528  0.00000
omega  -0.128127          NA         NA       NA

LogLikelihood : 12506.33

Information Criteria
------------------------------------

Akaike       -5.3586
Bayes        -5.3448
Shibata      -5.3586
Hannan-Quinn -5.3538

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
```

```
                          statistic p-value
Lag[1]                      0.02029  0.8867
Lag[2*(p+q)+(p+q)-1][11]   2.80160  1.0000
Lag[4*(p+q)+(p+q)-1][19]   5.93629  0.9732
d.o.f=4
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                        statistic p-value
Lag[1]                     0.3281  0.5668
Lag[2*(p+q)+(p+q)-1][5]    0.9595  0.8688
Lag[4*(p+q)+(p+q)-1][9]    2.0742  0.8960
d.o.f=2

Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[3]    0.1750 0.500 2.000  0.6757
ARCH Lag[5]    0.9116 1.440 1.667  0.7593
ARCH Lag[7]    1.3484 2.315 1.543  0.8512

Nyblom stability test
------------------------------------
Joint Statistic:  2.9285
Individual Statistics:
mu     0.1619
ar1    0.2230
ar2    0.1279
ma1    0.2270
ma2    0.1183
alpha1 0.4464
beta1  0.4648
gamma1 0.1543
skew   0.1428
shape  1.2140

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:         2.29 2.54 3.05
Individual Statistic:    0.35 0.47 0.75

Sign Bias Test
------------------------------------
```

```
                    t-value    prob sig
Sign Bias            2.1405 0.03237  **
Negative Sign Bias   0.8676 0.38565
Positive Sign Bias   2.0592 0.03953  **
Joint Effect         5.9424 0.11445


Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     30.10      0.05055
2    30     37.77      0.12761
3    40     45.97      0.20593
4    50     57.98      0.17797


Elapsed time : 1.785898
```

```r
VaR_EGARCH <- -(as.numeric(quantile(egarch11_fit, probs = t_couv)))

plot(
  as.zoo(cbind(rend_lvmh,-VaR_EGARCH)),
  screens=1,
  col = c("black","lightseagreen"),
  main="VaR ARMA-EGARCH(1,1)",
  xlab = "Temps",
  ylab = "Rendements"
)
points(as.zoo(rend_lvmh[as.numeric(rend_lvmh) < -VaR_EGARCH]), pch=20,
col="red")
legend("bottom",c("Rendements","VaR EGARCH", "Dépassements de la VaR"),
         col = c("black","lightseagreen", 'red'),lty = c(1,1,NA),lwd =
         c(2,2,NA),pch = c(NA,NA,20))
```




```
egarch11_estimation_fit <-
  rugarch::ugarchfit(
    egarch11_spec,
    rend_lvmh,
    out.sample = n_backtest
  )

forecast_egarch11 <-
  rugarch::ugarchforecast(
    egarch11_estimation_fit,
    data = rend_lvmh,
    n.roll = n_backtest-1,
    n.ahead = 1
  )

VaR_EGARCH_backtest <- -(as.numeric(quantile(forecast_egarch11, probs =
0.01)))

test_VaR_EGARCH<-
  rugarch::VaRTest(
    alpha = t_couv,
    actual = sample_backtest,
    VaR = -VaR_EGARCH_backtest,
```

```r
    conf.level = 0.95
  )

tibble(
  name = names(unlist(test_VaR_EGARCH)),
  value = unlist(test_VaR_EGARCH)
) |>
  print(n = 20)
```

```
# A tibble: 12 x 2
   name            value
   <chr>           <chr>
 1 expected.exceed 20
 2 actual.exceed   33
 3 uc.H0           Correct Exceedances
 4 uc.LRstat       7.13670995390553
 5 uc.critical     3.84145882069412
 6 uc.LRp          0.00755215918452179
 7 uc.Decision     Reject H0
 8 cc.H0           Correct Exceedances & Independent
 9 cc.LRstat       7.48366413009826
10 cc.critical     5.99146454710798
11 cc.LRp          0.0237106239097272
12 cc.Decision     Reject H0
```

```r
HIT <- if_else(sample_backtest < -VaR_EGARCH_backtest, 1, 0)

HIT_centre <- HIT - 0.01

stats::Box.test(HIT_centre, lag = 5, type = "Ljung-Box")
```

```
    Box-Ljung test

data:  HIT_centre
X-squared = 2.3334, df = 5, p-value = 0.8013
```

```r
df <-
  tibble(
    HIT_centre = HIT_centre,
    rend = sample_backtest,
    rend2 = sample_backtest**2
```

```r
  )

ARDL::ardl(
  HIT_centre ~ rend + rend2,
  data = df,
  order = 5
) |>
  summary()
```

```
Time series regression with "ts" data:
Start = 6, End = 2000

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
     Min       1Q   Median       3Q      Max
-0.60796 -0.02566  0.00439  0.02278  0.85526

Coefficients:
                  Estimate Std. Error t value            Pr(>|t|)
(Intercept)      -0.006437   0.003062  -2.102              0.0357 *
L(HIT_centre, 1) -0.006197   0.022501  -0.275              0.7830
L(HIT_centre, 2) -0.023666   0.022487  -1.052              0.2927
L(HIT_centre, 3)  0.021837   0.022457   0.972              0.3310
L(HIT_centre, 4)  0.001126   0.022444   0.050              0.9600
L(HIT_centre, 5)  0.002927   0.022082   0.133              0.8946
rend             -2.744581   0.127168 -21.582 <0.0000000000000002 ***
L(rend, 1)       -0.006977   0.141556  -0.049              0.9607
L(rend, 2)       -0.147915   0.141509  -1.045              0.2960
L(rend, 3)        0.006970   0.141487   0.049              0.9607
L(rend, 4)        0.050010   0.142374   0.351              0.7254
L(rend, 5)        0.027817   0.142132   0.196              0.8449
rend2            64.778756   2.808040  23.069 <0.0000000000000002 ***
L(rend2, 1)      -1.501057   3.159774  -0.475              0.6348
L(rend2, 2)      -5.763417   3.135988  -1.838              0.0662 .
L(rend2, 3)      -8.022119   3.132232  -2.561              0.0105 *
L(rend2, 4)      -4.552029   3.144437  -1.448              0.1479
L(rend2, 5)      -6.167844   3.140954  -1.964              0.0497 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.1041 on 1977 degrees of freedom
Multiple R-squared:  0.3195,    Adjusted R-squared:  0.3136
F-statistic:  54.6 on 17 and 1977 DF,  p-value: < 0.00000000000000022
```

```r
egarch22_spec <-
  rugarch::ugarchspec(
    variance.model =
      list(
        model = "eGARCH",
        garchOrder = c(2,2),
        variance.targeting = TRUE
      ),
    mean.model =
      list(
        armaOrder = c(ar,ma),
        include.mean = TRUE
      ),
    distribution.model = "sged"
  )

egarch22_fit <-
  rugarch::ugarchfit(
    egarch22_spec,
    rend_lvmh
  )

egarch22_fit
```

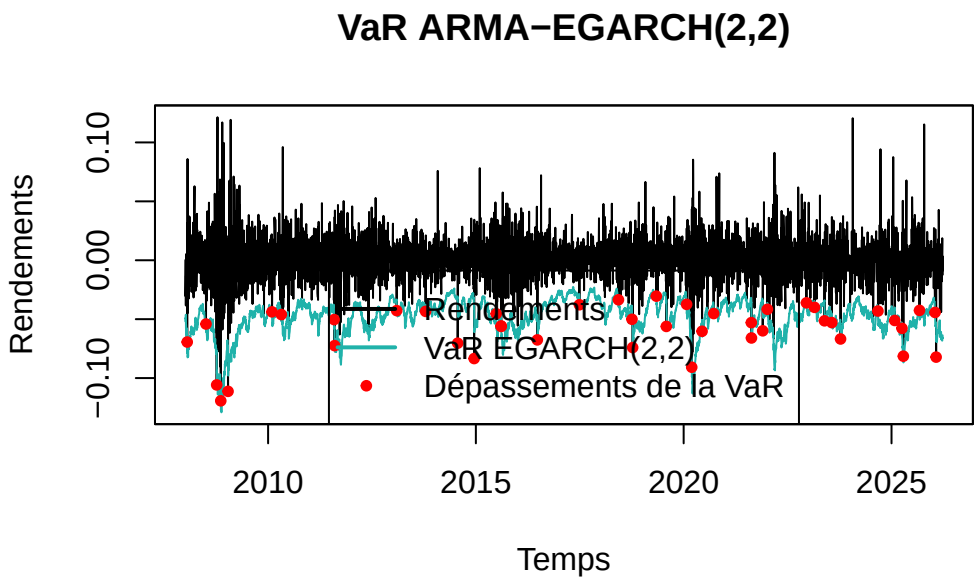
```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model : eGARCH(2,2)
Mean Model  : ARFIMA(2,0,2)
Distribution    : sged

Optimal Parameters
------------------------------------
        Estimate  Std. Error   t value Pr(>|t|)
```

```
mu      0.000166    0.000161    1.0298 0.303087
ar1     1.152698    0.040561   28.4190 0.000000
ar2    -0.635380    0.016949  -37.4876 0.000000
ma1    -1.168654    0.042377  -27.5777 0.000000
ma2     0.624816    0.016101   38.8059 0.000000
alpha1 -0.073905    0.011362   -6.5047 0.000000
alpha2 -0.083806    0.013096   -6.3993 0.000000
beta1   0.072493    0.000297  243.8437 0.000000
beta2   0.896539    0.000834 1074.4119 0.000000
gamma1  0.090616    0.017747    5.1061 0.000000
gamma2  0.069303    0.019851    3.4911 0.000481
skew    0.972333    0.016651   58.3952 0.000000
shape   1.277628    0.034442   37.0950 0.000000
omega  -0.246696          NA        NA       NA

Robust Standard Errors:
        Estimate  Std. Error   t value Pr(>|t|)
mu      0.000166    0.000165    1.0107 0.312145
ar1     1.152698    0.012787   90.1484 0.000000
ar2    -0.635380    0.005304 -119.8028 0.000000
ma1    -1.168654    0.013968  -83.6643 0.000000
ma2     0.624816    0.005764  108.3966 0.000000
alpha1 -0.073905    0.011687   -6.3234 0.000000
alpha2 -0.083806    0.012844   -6.5250 0.000000
beta1   0.072493    0.002368   30.6173 0.000000
beta2   0.896539    0.000993  902.4382 0.000000
gamma1  0.090616    0.016184    5.5990 0.000000
gamma2  0.069303    0.018986    3.6503 0.000262
skew    0.972333    0.018233   53.3283 0.000000
shape   1.277628    0.042616   29.9801 0.000000
omega  -0.246696          NA        NA       NA

LogLikelihood : 12506.6

Information Criteria
------------------------------------

Akaike       -5.3575
Bayes        -5.3395
Shibata      -5.3575
Hannan-Quinn -5.3511

Weighted Ljung-Box Test on Standardized Residuals
```

```
------------------------------------
                         statistic p-value
Lag[1]                     0.02688  0.8698
Lag[2*(p+q)+(p+q)-1][11]   2.71825  1.0000
Lag[4*(p+q)+(p+q)-1][19]   5.87793  0.9753
d.o.f=4
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                         statistic p-value
Lag[1]                      0.2096  0.6471
Lag[2*(p+q)+(p+q)-1][11]    3.4183  0.8158
Lag[4*(p+q)+(p+q)-1][19]    7.9264  0.6840
d.o.f=4

Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[5]     1.382 0.500 2.000  0.2398
ARCH Lag[7]     1.745 1.473 1.746  0.5625
ARCH Lag[9]     3.054 2.402 1.619  0.5555

Nyblom stability test
------------------------------------
Joint Statistic:  3.3196
Individual Statistics:
mu     0.16211
ar1    0.21629
ar2    0.12114
ma1    0.21942
ma2    0.11220
alpha1 0.32670
alpha2 0.25655
beta1  0.48361
beta2  0.45825
gamma1 0.03889
gamma2 0.44432
skew   0.14195
shape  1.20548

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:         2.89 3.15 3.69
```

```
Individual Statistic:    0.35 0.47 0.75

Sign Bias Test
------------------------------------
                   t-value    prob sig
Sign Bias           2.1019 0.03562  **
Negative Sign Bias  0.8537 0.39330
Positive Sign Bias  2.2415 0.02504  **
Joint Effect        6.3973 0.09380   *


Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     31.75      0.03330
2    30     40.48      0.07637
3    40     50.13      0.10910
4    50     61.69      0.10544


Elapsed time : 3.421628
```

```r
VaR_EGARCH22 <- -(as.numeric(quantile(egarch22_fit, probs = t_couv)))

plot(
  as.zoo(cbind(rend_lvmh,-VaR_EGARCH22)),
  screens=1,
  col = c("black","lightseagreen"),
  main="VaR ARMA-EGARCH(2,2)",
  xlab = "Temps",
  ylab = "Rendements"
)
points(as.zoo(rend_lvmh[as.numeric(rend_lvmh) < -VaR_EGARCH22]), pch=20,
col="red")
legend("bottom",c("Rendements","VaR EGARCH(2,2)", "Dépassements de la VaR"),
         col = c("black","lightseagreen", 'red'),lty = c(1,1,NA),lwd =
         c(2,2,NA),pch = c(NA,NA,20))
```




```
egarch22_estimation_fit <-
  rugarch::ugarchfit(
    egarch22_spec,
    rend_lvmh,
    out.sample = n_backtest
  )

forecast_egarch22 <-
  rugarch::ugarchforecast(
    egarch22_estimation_fit,
    data = rend_lvmh,
    n.roll = n_backtest-1,
    n.ahead = 1
  )

VaR_EGARCH22_backtest <- -(as.numeric(quantile(forecast_egarch22, probs =
0.01)))

test_VaR_EGARCH22<-
  rugarch::VaRTest(
    alpha = t_couv,
    actual = sample_backtest,
    VaR = -VaR_EGARCH22_backtest,
```

```r
    conf.level = 0.95
  )

tibble(
  name = names(unlist(test_VaR_EGARCH22)),
  value = unlist(test_VaR_EGARCH22)
) |>
  print(n = 20)
```

```
# A tibble: 12 x 2
   name            value
   <chr>           <chr>
 1 expected.exceed 20
 2 actual.exceed   33
 3 uc.H0           Correct Exceedances
 4 uc.LRstat       7.13670995390553
 5 uc.critical     3.84145882069412
 6 uc.LRp          0.00755215918452179
 7 uc.Decision     Reject H0
 8 cc.H0           Correct Exceedances & Independent
 9 cc.LRstat       13.0095513372679
10 cc.critical     5.99146454710798
11 cc.LRp          0.00149627638279803
12 cc.Decision     Reject H0
```

```r
HIT <- if_else(sample_backtest < -VaR_EGARCH22_backtest, 1, 0)

HIT_centre <- HIT - 0.01

stats::Box.test(HIT_centre, lag = 5, type = "Ljung-Box")
```

```
    Box-Ljung test

data:  HIT_centre
X-squared = 13.528, df = 5, p-value = 0.01891
```

```r
df <-
  tibble(
    HIT_centre = HIT_centre,
    rend = sample_backtest,
    rend2 = sample_backtest**2
```

```r
  )

ARDL::ardl(
  HIT_centre ~ rend + rend2,
  data = df,
  order = 5
) |>
  summary()
```

```
Time series regression with "ts" data:
Start = 6, End = 2000

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
     Min       1Q   Median       3Q      Max
-0.59603 -0.02628  0.00448  0.02283  0.85761

Coefficients:
                  Estimate Std. Error t value             Pr(>|t|)
(Intercept)      -0.006077   0.003072  -1.978              0.04807 *
L(HIT_centre, 1)  0.067740   0.022488   3.012              0.00263 **
L(HIT_centre, 2) -0.029915   0.022525  -1.328              0.18431
L(HIT_centre, 3)  0.021857   0.022492   0.972              0.33129
L(HIT_centre, 4) -0.002501   0.022498  -0.111              0.91149
L(HIT_centre, 5) -0.007926   0.022080  -0.359              0.71966
rend             -2.707470   0.127641 -21.212 < 0.0000000000000002 ***
L(rend, 1)        0.049106   0.141556   0.347              0.72870
L(rend, 2)       -0.103389   0.141398  -0.731              0.46475
L(rend, 3)       -0.059410   0.141305  -0.420              0.67421
L(rend, 4)        0.107226   0.142203   0.754              0.45092
L(rend, 5)       -0.048209   0.142094  -0.339              0.73444
rend2            63.585389   2.817449  22.568 < 0.0000000000000002 ***
L(rend2, 1)      -3.362464   3.154567  -1.066              0.28660
L(rend2, 2)      -5.519617   3.129884  -1.764              0.07797 .
L(rend2, 3)      -8.235353   3.125705  -2.635              0.00849 **
L(rend2, 4)      -4.508706   3.138587  -1.437              0.15101
L(rend2, 5)      -5.205393   3.136196  -1.660              0.09712 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.1045 on 1977 degrees of freedom
Multiple R-squared:  0.3147,    Adjusted R-squared:  0.3088
F-statistic:  53.4 on 17 and 1977 DF,  p-value: < 0.00000000000000022
```

```r
df_VaR <-
  tibble(
    VaR_VCov = VaR_VCov,
    VaR_CFisher = VaR_CFisher,
    VaR_student = VaR_student,
    VaR_GARCH= VaR_GARCH,
    VaR_EGARCH = VaR_EGARCH,
    VaR_EGARCH22 = VaR_EGARCH22
  ) |>
  drop_na()

df_VaR_long <-
  df_VaR |>
  mutate(temps = row_number()) |>
  pivot_longer(
    cols = -temps,
    names_to = "model",
    values_to = "VaR"
  ) |>
  mutate(
    .by = temps,
    moyenne_VaR = mean(VaR)
  ) |>
  mutate(
    ecart = (VaR - moyenne_VaR) / moyenne_VaR
  ) |>
  summarize(
    .by = model,
    MRB = abs(mean(ecart))
  ) |>
  print()
```

```
# A tibble: 6 x 2
  model          MRB
  <chr>        <dbl>
1 VaR_VCov     0.251
2 VaR_CFisher  0.301
3 VaR_student  0.546
```

```
4 VaR_GARCH    0.204
5 VaR_EGARCH   0.196
6 VaR_EGARCH22 0.196
```

Exercice 4 : Dow Jones

Quel modèle de VaR décidez-vous de prendre ? Est-il un bon modèle de VaR ? Si non comment l'améliorer ?

```r
rm(list = ls())
source(here::here("02-codes", "utils", "setup.R"))
```

```r
dj <-
  pdfetch::pdfetch_YAHOO(
    "RMS.PA",
    fields = "adjclose",
    from = as.Date("2007-01-03"),
    to = as.Date("2026-03-30")
  )

rend_dj <-
  na.omit((log(dj) - log(lag(dj))) * 100)

ar <- 6
integration <- 0
ma <- 0

t_couv <- 0.01

n_backtest <- 2000

taille_sample_estimation <- length(rend_dj) - n_backtest

sample_estimation <- rend_dj[1:taille_sample_estimation]

sample_backtest <- rend_dj[(taille_sample_estimation+1):length(rend_dj)]
```

```r
garch11_spec <-
  rugarch::ugarchspec(
    variance.model =
```

```r
      list(
        model = "sGARCH",
        garchOrder = c(1,1),
        variance.targeting = FALSE
      ),
    mean.model =
      list(
        armaOrder = c(ar,ma),
        include.mean = FALSE
      ),
    distribution.model = "norm"
  )
```

```r
garch11_fit <-
  rugarch::ugarchfit(
    garch11_spec,
    rend_dj
  )

garch11_fit
```

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model : sGARCH(1,1)
Mean Model  : ARFIMA(6,0,0)
Distribution    : norm

Optimal Parameters
------------------------------------
        Estimate  Std. Error   t value Pr(>|t|)
ar1     0.005362    0.015151  0.353915 0.723403
ar2     0.013336    0.015260  0.873876 0.382186
ar3    -0.021472    0.015210 -1.411726 0.158031
ar4     0.010115    0.015200  0.665479 0.505744
ar5     0.000224    0.015094  0.014819 0.988177
ar6    -0.026167    0.015111 -1.731656 0.083335
omega   0.029887    0.008736  3.421103 0.000624
```

```
alpha1  0.059906    0.009329  6.421248 0.000000
beta1   0.931988    0.011044 84.386300 0.000000

Robust Standard Errors:
        Estimate  Std. Error   t value Pr(>|t|)
ar1     0.005362    0.016713  0.320850 0.748324
ar2     0.013336    0.015662  0.851431 0.394530
ar3    -0.021472    0.016989 -1.263864 0.206279
ar4     0.010115    0.015509  0.652215 0.514262
ar5     0.000224    0.015817  0.014141 0.988717
ar6    -0.026167    0.015766 -1.659674 0.096980
omega   0.029887    0.022330  1.338406 0.180764
alpha1  0.059906    0.026964  2.221697 0.026304
beta1   0.931988    0.031475 29.610321 0.000000

LogLikelihood : -9176.04

Information Criteria
------------------------------------

Akaike       3.7338
Bayes        3.7456
Shibata      3.7337
Hannan-Quinn 3.7379

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic p-value
Lag[1]                     0.1234  0.7254
Lag[2*(p+q)+(p+q)-1][17]    8.2698  0.8928
Lag[4*(p+q)+(p+q)-1][29]   13.3242  0.7048
d.o.f=6
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                        statistic   p-value
Lag[1]                      14.31 0.0001549
Lag[2*(p+q)+(p+q)-1][5]     14.59 0.0005719
Lag[4*(p+q)+(p+q)-1][9]     15.48 0.0027281
d.o.f=2

Weighted ARCH LM Tests
```

```
------------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[3]    0.1312 0.500 2.000  0.7172
ARCH Lag[5]    0.6310 1.440 1.667  0.8444
ARCH Lag[7]    1.3835 2.315 1.543  0.8444

Nyblom stability test
------------------------------------
Joint Statistic:  1.5485
Individual Statistics:
ar1    0.1646
ar2    0.1151
ar3    0.1519
ar4    0.2054
ar5    0.2081
ar6    0.1301
omega  0.1706
alpha1 0.1858
beta1  0.1564

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:         2.1 2.32 2.82
Individual Statistic:    0.35 0.47 0.75

Sign Bias Test
------------------------------------
                   t-value    prob sig
Sign Bias           0.7616 0.44636
Negative Sign Bias  2.5312 0.01140  **
Positive Sign Bias  1.6527 0.09845   *
Joint Effect        9.2477 0.02617  **


Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic                      p-value(g-1)
1    20     170.4 0.000000000000000000000000023572
2    30     192.8 0.000000000000000000000000042066
3    40     218.4 0.000000000000000000000000008332
4    50     238.7 0.000000000000000000000000009252


Elapsed time : 0.319622
```

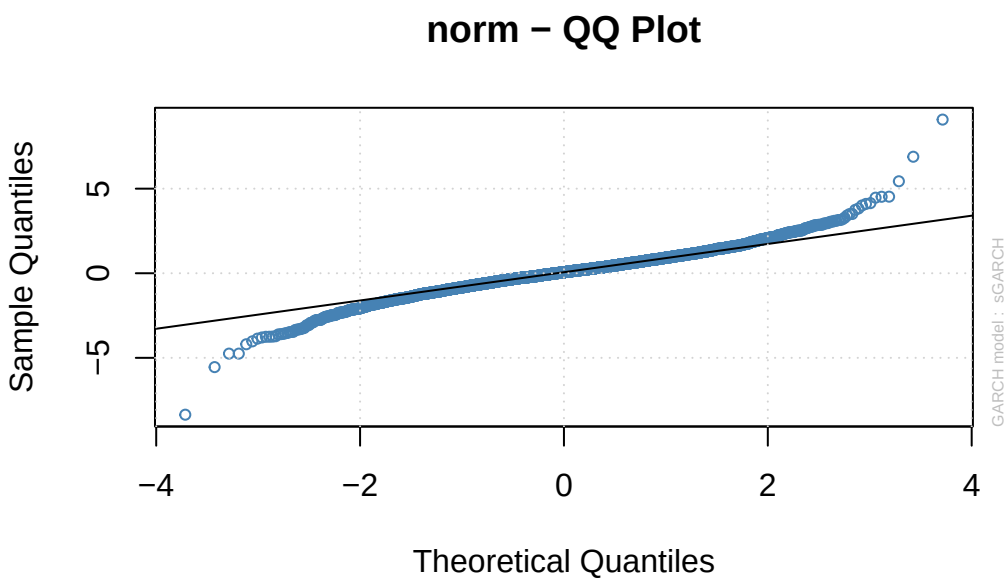
```r
rugarch::persistence(garch11_fit)
```

```
[1] 0.9918943
```

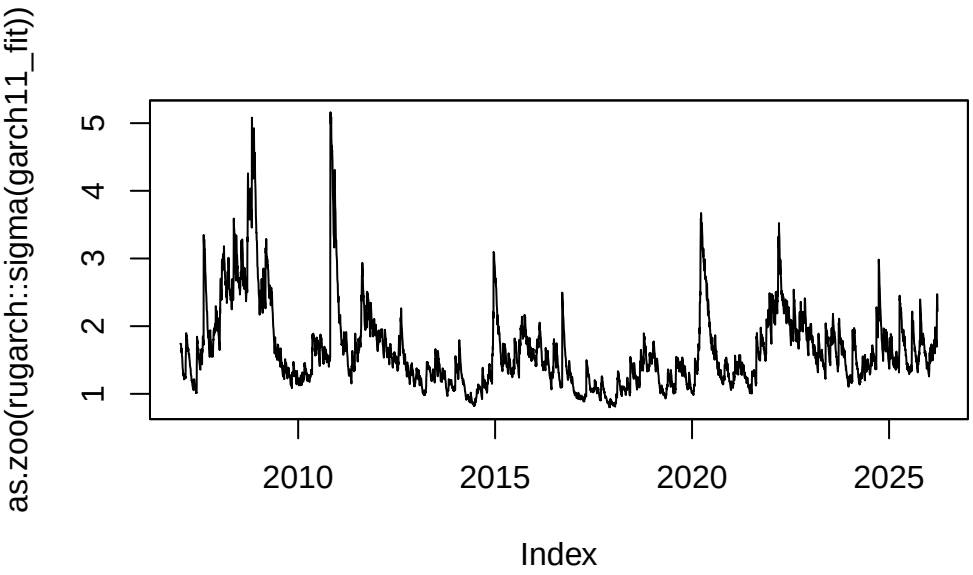
```r
rugarch::halflife(garch11_fit)
```

```
[1] 85.16626
```

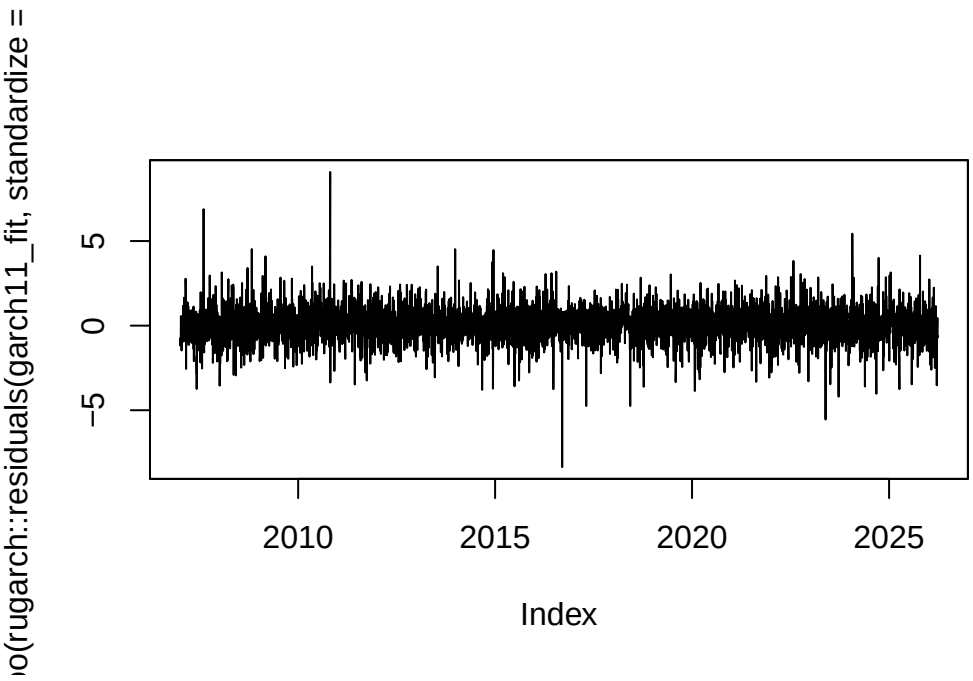
```r
plot(garch11_fit, which = 9)
```




```r
rugarch::sigma(garch11_fit) |>
  as.zoo() |>
  plot()
```




```r
rugarch::residuals(garch11_fit, standardize = TRUE) |>
  as.zoo() |>
  plot()
```



```r
egarch11_spec <-
  rugarch::ugarchspec(
    variance.model =
      list(
        model = "eGARCH",
        garchOrder = c(1,1),
        variance.targeting = FALSE
      ),
    mean.model =
      list(
        armaOrder = c(ar,ma),
        include.mean = FALSE
      ),
    distribution.model = "sged"
  )
```

```r
egarch11_fit <-
  rugarch::ugarchfit(
    egarch11_spec,
    rend_dj
  )

egarch11_fit
```

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model : eGARCH(1,1)
Mean Model  : ARFIMA(6,0,0)
Distribution    : sged

Optimal Parameters
------------------------------------
        Estimate  Std. Error   t value Pr(>|t|)
ar1    -0.004392    0.008378  -0.52431 0.600065
ar2     0.006640    0.012626   0.52589 0.598967
ar3    -0.021164    0.011359  -1.86320 0.062434
ar4     0.027912    0.012525   2.22850 0.025847
```

```
ar5     0.009133    0.012907   0.70755 0.479223
ar6    -0.011594    0.010789  -1.07464 0.282537
omega   0.012697    0.001571   8.07957 0.000000
alpha1 -0.028760    0.007707  -3.73137 0.000190
beta1   0.987429    0.001177 838.93331 0.000000
gamma1  0.139639    0.019858   7.03185 0.000000
skew    0.943621    0.014118  66.83630 0.000000
shape   1.244939    0.031963  38.94893 0.000000

Robust Standard Errors:
        Estimate  Std. Error   t value Pr(>|t|)
ar1    -0.004392    0.008289  -0.52990 0.596184
ar2     0.006640    0.015650   0.42426 0.671375
ar3    -0.021164    0.014930  -1.41756 0.156319
ar4     0.027912    0.014469   1.92915 0.053712
ar5     0.009133    0.016680   0.54751 0.584029
ar6    -0.011594    0.013849  -0.83719 0.402488
omega   0.012697    0.002128   5.96517 0.000000
alpha1 -0.028760    0.010539  -2.72899 0.006353
beta1   0.987429    0.002750 359.09405 0.000000
gamma1  0.139639    0.037674   3.70652 0.000210
skew    0.943621    0.017250  54.70208 0.000000
shape   1.244939    0.045002  27.66391 0.000000

LogLikelihood : -8961.185

Information Criteria
------------------------------------

Akaike       3.6476
Bayes        3.6635
Shibata      3.6476
Hannan-Quinn 3.6532

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic    p-value
Lag[1]                      1.033 0.30945290
Lag[2*(p+q)+(p+q)-1][17]   11.450 0.00006649
Lag[4*(p+q)+(p+q)-1][29]   17.507 0.18750663
d.o.f=6
H0 : No serial correlation
```

```
Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                        statistic           p-value
Lag[1]                      43.62 0.000000000039970
Lag[2*(p+q)+(p+q)-1][5]     44.46 0.000000000002447
Lag[4*(p+q)+(p+q)-1][9]     45.43 0.000000000020914
d.o.f=2

Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[3]    0.5683 0.500 2.000  0.4509
ARCH Lag[5]    0.8493 1.440 1.667  0.7781
ARCH Lag[7]    1.7008 2.315 1.543  0.7803

Nyblom stability test
------------------------------------
Joint Statistic:  3.9691
Individual Statistics:
ar1    0.10054
ar2    0.19298
ar3    0.06585
ar4    0.25213
ar5    0.07435
ar6    0.10103
omega  0.24206
alpha1 0.30330
beta1  0.35348
gamma1 0.77195
skew   1.66432
shape  0.07803

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:         2.69 2.96 3.51
Individual Statistic:    0.35 0.47 0.75

Sign Bias Test
------------------------------------
                   t-value      prob sig
Sign Bias           0.6175 0.5369041
Negative Sign Bias  2.4186 0.0156156  **
Positive Sign Bias  3.1833 0.0014650 ***
Joint Effect       17.5046 0.0005564 ***
```

```
Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     38.89     0.004569
2    30     42.12     0.054757
3    40     58.89     0.021328
4    50     68.07     0.036994


Elapsed time : 2.43912
```

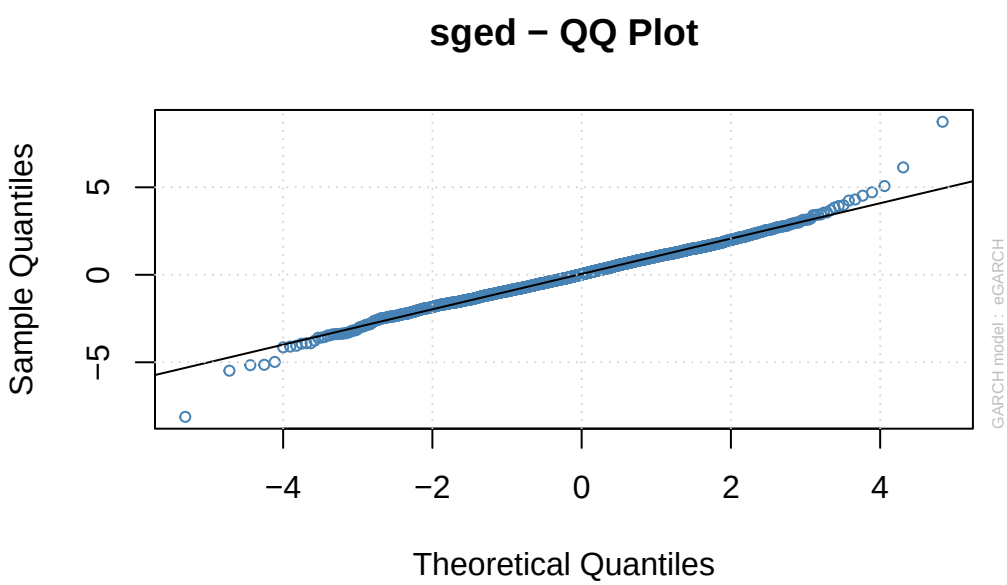
```r
rugarch::persistence(egarch11_fit)
```

```
[1] 0.9874291
```

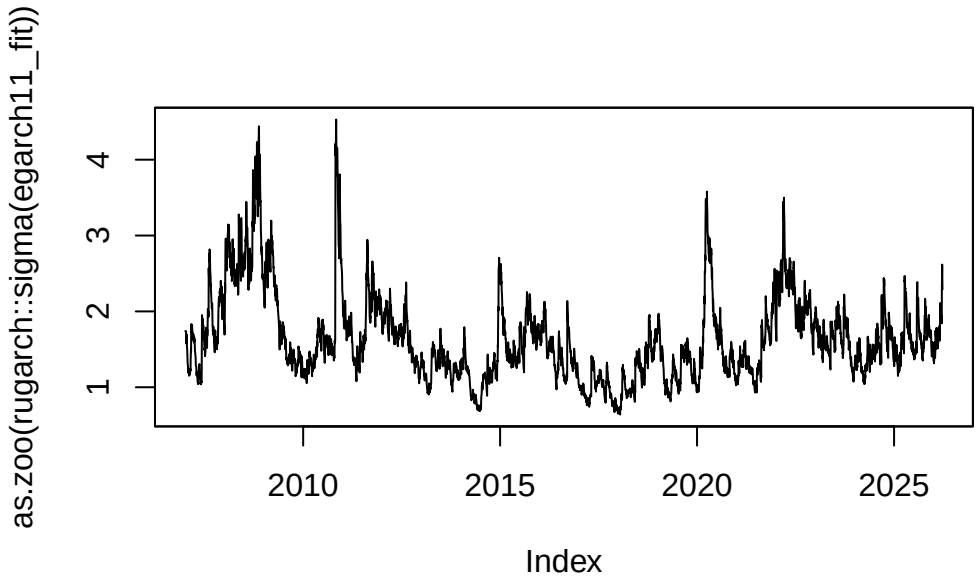
```r
rugarch::halflife(egarch11_fit)
```

```
[1] 54.79151
```

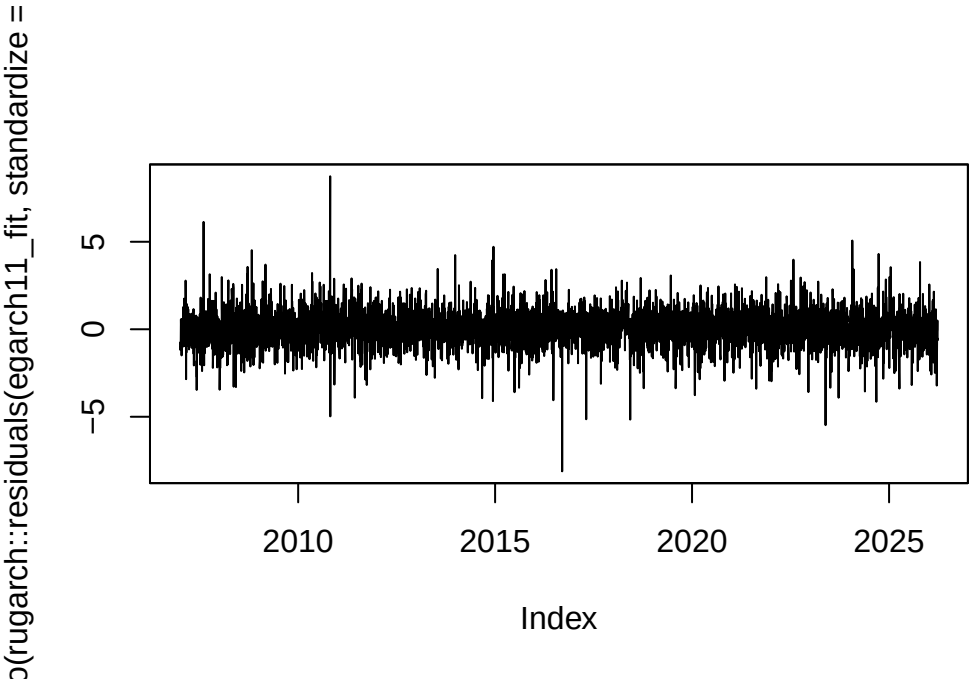
```r
plot(egarch11_fit, which = 9)
```



```r
rugarch::sigma(egarch11_fit) |>
  as.zoo() |>
  plot()
```




```r
rugarch::residuals(egarch11_fit, standardize = TRUE) |>
  as.zoo() |>
  plot()
```




```
gjr_garch11_spec <-
  rugarch::ugarchspec(
    variance.model =
      list(
        model = "gjrGARCH",
        garchOrder = c(1,1),
        variance.targeting = FALSE
      ),
    mean.model =
      list(
        armaOrder = c(ar,ma),
        include.mean = FALSE
      ),
    distribution.model = "sstd"
  )
```

```
gjr_garch11_fit <-
  rugarch::ugarchfit(
    gjr_garch11_spec,
    rend_dj
  )

gjr_garch11_fit
```

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model : gjrGARCH(1,1)
Mean Model  : ARFIMA(6,0,0)
Distribution    : sstd

Optimal Parameters
------------------------------------
        Estimate  Std. Error   t value Pr(>|t|)
ar1     0.001214    0.014268  0.085111 0.932173
ar2     0.009295    0.013886  0.669344 0.503276
ar3    -0.020261    0.013813 -1.466830 0.142422
ar4     0.022592    0.013837  1.632754 0.102521
ar5     0.006738    0.013678  0.492643 0.622265
ar6    -0.008724    0.013681 -0.637706 0.523665
omega   0.019525    0.007007  2.786429 0.005329
alpha1  0.034926    0.008585  4.068147 0.000047
beta1   0.943507    0.010126 93.176598 0.000000
gamma1  0.038022    0.010653  3.569135 0.000358
skew    0.944508    0.016949 55.726025 0.000000
shape   5.097886    0.375132 13.589572 0.000000

Robust Standard Errors:
        Estimate  Std. Error   t value Pr(>|t|)
ar1     0.001214    0.015485  0.078422 0.937492
ar2     0.009295    0.014216  0.653799 0.513241
ar3    -0.020261    0.013559 -1.494280 0.135102
ar4     0.022592    0.014351  1.574231 0.115434
ar5     0.006738    0.014092  0.478163 0.632534
ar6    -0.008724    0.014321 -0.609206 0.542388
omega   0.019525    0.011743  1.662656 0.096381
alpha1  0.034926    0.013811  2.528848 0.011444
beta1   0.943507    0.018441 51.164562 0.000000
gamma1  0.038022    0.014884  2.554575 0.010632
skew    0.944508    0.017763 53.173755 0.000000
shape   5.097886    0.397021 12.840343 0.000000
```

```
LogLikelihood : -8946.287

Information Criteria
------------------------------------

Akaike       3.6416
Bayes        3.6574
Shibata      3.6416
Hannan-Quinn 3.6471

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic p-value
Lag[1]                     0.2991  0.5845
Lag[2*(p+q)+(p+q)-1][17]  10.3159  0.0163
Lag[4*(p+q)+(p+q)-1][29]  16.0036  0.3432
d.o.f=6
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                        statistic      p-value
Lag[1]                      26.00 0.0000003414
Lag[2*(p+q)+(p+q)-1][5]     26.45 0.0000003131
Lag[4*(p+q)+(p+q)-1][9]     27.22 0.0000024797
d.o.f=2

Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[3]    0.4830 0.500 2.000  0.4871
ARCH Lag[5]    0.6576 1.440 1.667  0.8363
ARCH Lag[7]    1.3381 2.315 1.543  0.8532

Nyblom stability test
------------------------------------
Joint Statistic:  4.5653
Individual Statistics:
ar1    0.07624
ar2    0.18904
ar3    0.07893
ar4    0.29244
ar5    0.07328
```

```
ar6    0.06194
omega  0.31535
alpha1 0.42387
beta1  0.29740
gamma1 0.25055
skew   1.61645
shape  0.33796

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:         2.69 2.96 3.51
Individual Statistic:    0.35 0.47 0.75

Sign Bias Test
------------------------------------
                   t-value     prob sig
Sign Bias            0.578 0.563279
Negative Sign Bias   1.938 0.052648   *
Positive Sign Bias   2.581 0.009869 ***
Joint Effect        11.656 0.008660 ***


Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     23.11       0.2327
2    30     40.89       0.0704
3    40     36.46       0.5865
4    50     59.33       0.1482


Elapsed time : 1.014454
```

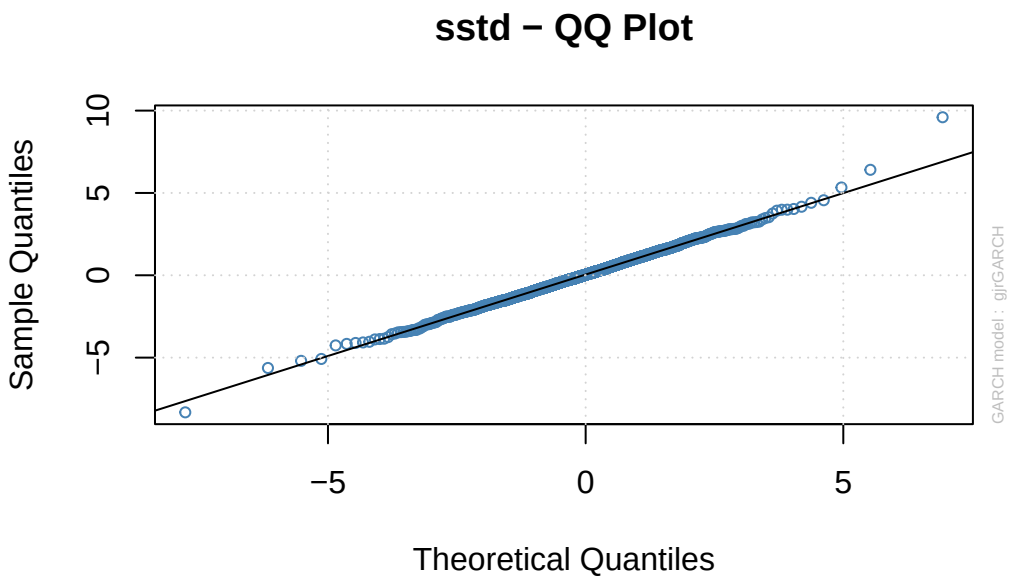
```r
rugarch::persistence(gjr_garch11_fit)
```

```
[1] 0.9969736
```

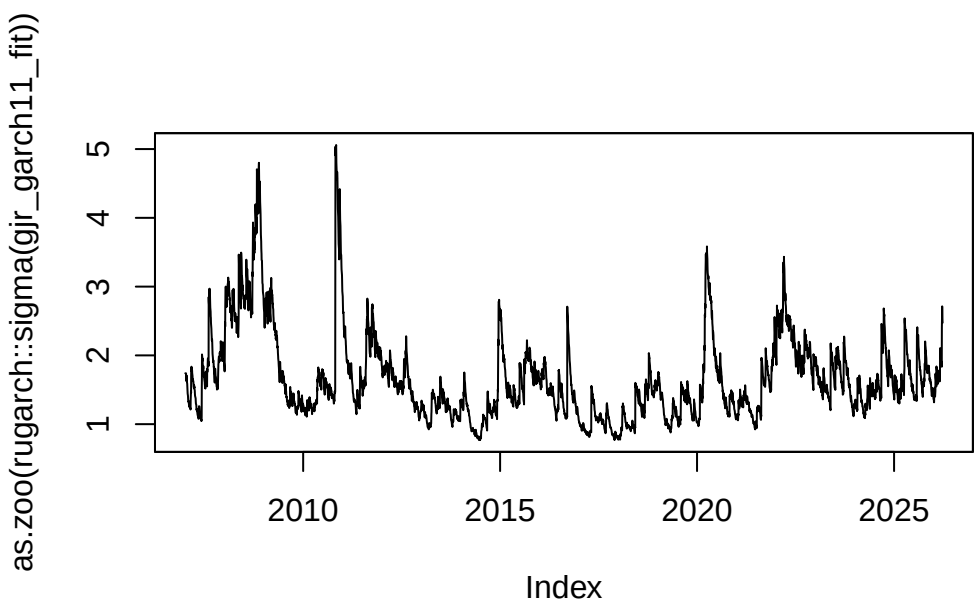
```r
rugarch::halflife(gjr_garch11_fit)
```

```
[1] 228.6837
```

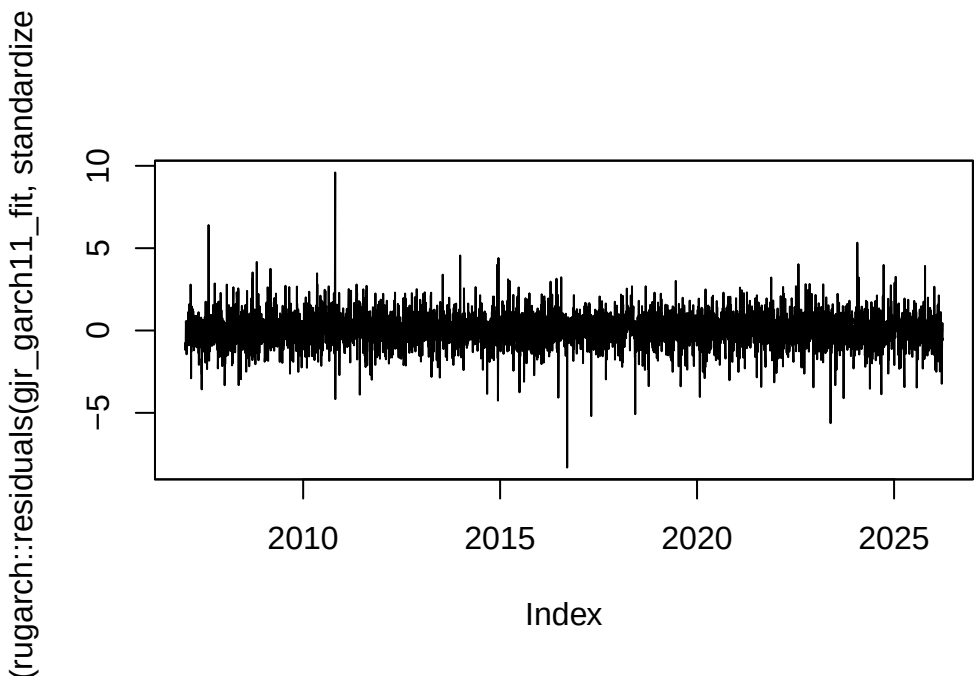
```r
plot(gjr_garch11_fit, which = 9)
```




```
rugarch::sigma(gjr_garch11_fit) |>
  as.zoo() |>
  plot()
```



```r
rugarch::residuals(gjr_garch11_fit, standardize = TRUE) |>
  as.zoo() |>
  plot()
```



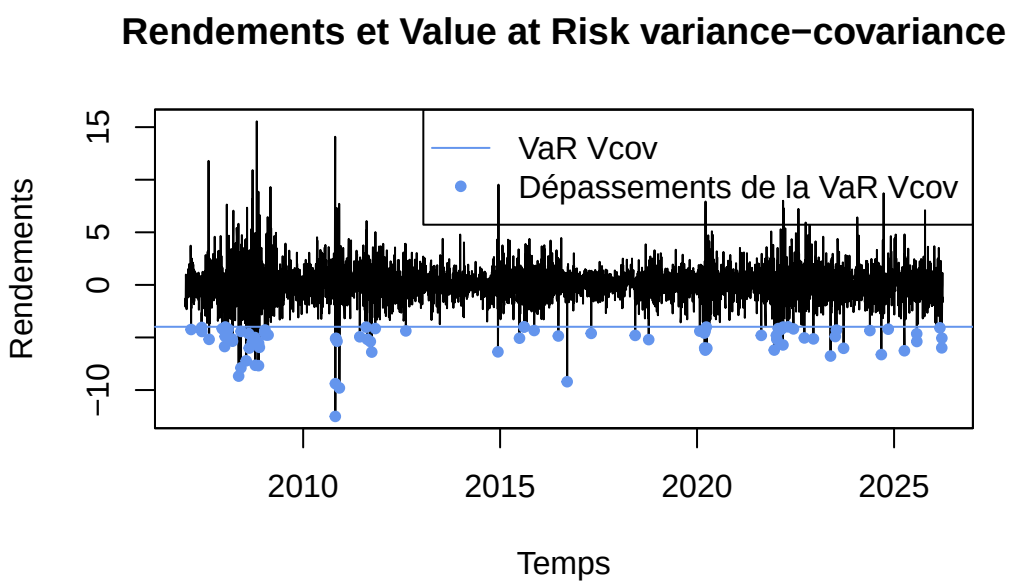

```r
tibble(
  criterion = c("Akaike", "Bayes", "Shibata", "Hannan-Quinn"),
  garch_11 = rugarch::infocriteria(garch11_fit) |> as.numeric(),
  egarch_11 = rugarch::infocriteria(egarch11_fit) |> as.numeric(),
  gjr_garch_11 = rugarch::infocriteria(gjr_garch11_fit) |> as.numeric()
)
```

```
# A tibble: 4 x 4
  criterion    garch_11 egarch_11 gjr_garch_11
  <chr>           <dbl>     <dbl>        <dbl>
1 Akaike           3.73      3.65         3.64
2 Bayes            3.75      3.66         3.66
3 Shibata          3.73      3.65         3.64
4 Hannan-Quinn     3.74      3.65         3.65
```

```r
VaR_VCov <- (-(mean(rend_dj)+sd(rend_dj)*qnorm(t_couv))) |> print()
```

```
[1] 3.98342
```

```r
plot(as.zoo(rend_dj),
      main = "Rendements et Value at Risk variance-covariance",
      xlab = "Temps",
      ylab = "Rendements")
  abline(h= -VaR_VCov, col="cornflowerblue")
  points(as.zoo(rend_dj[as.numeric(rend_dj) < -VaR_VCov]), pch=20,
  col="cornflowerblue")
  legend("topright", legend = c("VaR Vcov","Dépassements de la VaR Vcov"),
         pch = c(NA,20), lty = c(1, NA), col = "cornflowerblue")
```




```r
VaR_VCov_estimation <-
(-(mean(sample_estimation)+sd(sample_estimation)*qnorm(t_couv)))

test_VaR_VCov <-
  rugarch::VaRTest(
    alpha = t_couv,
    actual = sample_backtest,
    VaR = rep(-VaR_VCov_estimation, times = length(sample_backtest)),
    conf.level = 0.95
  )

tibble(
```

```r
  name = names(unlist(test_VaR_VCov)),
  value = unlist(test_VaR_VCov)
) |>
  print(n = 20)
```

```
# A tibble: 12 x 2
   name            value
   <chr>           <chr>
 1 expected.exceed 20
 2 actual.exceed   29
 3 uc.H0           Correct Exceedances
 4 uc.LRstat       3.59165748872128
 5 uc.critical     3.84145882069412
 6 uc.LRp          0.0580702969680549
 7 uc.Decision     Fail to Reject H0
 8 cc.H0           Correct Exceedances & Independent
 9 cc.LRstat       6.8480438258995
10 cc.critical     5.99146454710798
11 cc.LRp          0.0325811325936156
12 cc.Decision     Reject H0
```

```r
HIT_VCov <-
  if_else(sample_backtest < -VaR_VCov_estimation, 1, 0)

HIT_centre_VCov <- HIT_VCov - t_couv

stats::Box.test(HIT_centre_VCov, lag = 5, type = "Ljung-Box")
```

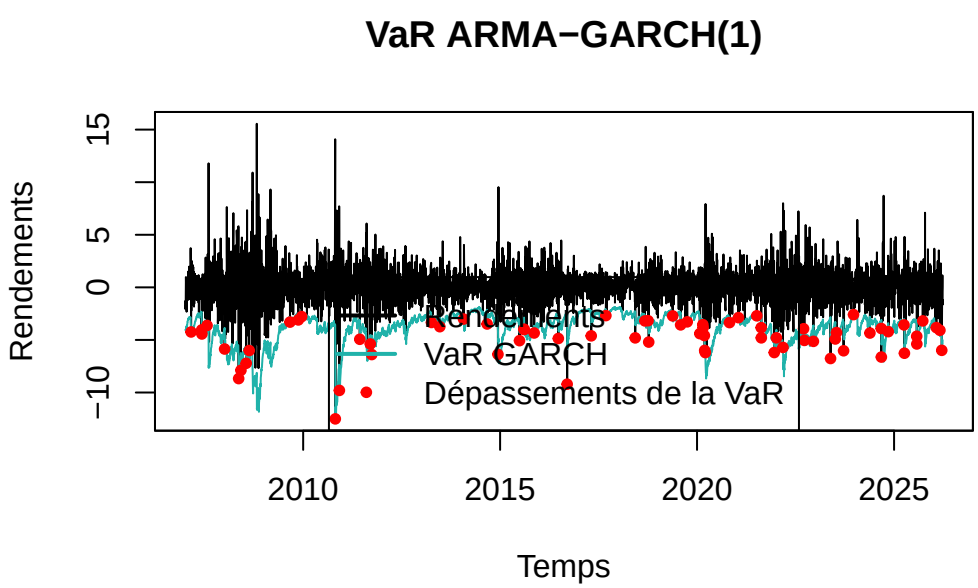
```
    Box-Ljung test

data:  HIT_centre_VCov
X-squared = 24.903, df = 5, p-value = 0.0001454
```

```r
df <- tibble(
  HIT_centre = HIT_centre_VCov,
  rend = sample_backtest |> as.numeric(),
  rend2 = sample_backtest**2 |> as.numeric()
)

ARDL::ardl(
  HIT_centre ~ rend + rend2,
```

```r
  data = df,
  order = 5
) |>
  summary()
```

```
Time series regression with "ts" data:
Start = 6, End = 2000

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
     Min       1Q   Median       3Q      Max
-0.50093 -0.01874  0.01199  0.02519  0.72608

Coefficients:
                   Estimate Std. Error t value             Pr(>|t|)
(Intercept)      -0.0155571  0.0029347  -5.301          0.000000128 ***
L(HIT_centre, 1)  0.0050220  0.0224227   0.224               0.8228
L(HIT_centre, 2)  0.0288762  0.0224375   1.287               0.1983
L(HIT_centre, 3)  0.0205332  0.0224528   0.915               0.3606
L(HIT_centre, 4)  0.0160389  0.0224435   0.715               0.4749
L(HIT_centre, 5)  0.0173547  0.0224155   0.774               0.4389
rend             -0.0284916  0.0012357 -23.056 < 0.0000000000000002 ***
L(rend, 1)       -0.0003255  0.0013931  -0.234               0.8153
L(rend, 2)       -0.0021633  0.0013901  -1.556               0.1198
L(rend, 3)        0.0013251  0.0013934   0.951               0.3417
L(rend, 4)        0.0021089  0.0013947   1.512               0.1307
L(rend, 5)       -0.0027652  0.0013961  -1.981               0.0478 *
rend2             0.0106742  0.0003546  30.101 < 0.0000000000000002 ***
L(rend2, 1)      -0.0008735  0.0004265  -2.048               0.0407 *
L(rend2, 2)      -0.0005035  0.0004262  -1.182               0.2375
L(rend2, 3)      -0.0003604  0.0004256  -0.847               0.3972
L(rend2, 4)      -0.0002577  0.0004250  -0.606               0.5443
L(rend2, 5)      -0.0009504  0.0004221  -2.252               0.0244 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.0913 on 1977 degrees of freedom
Multiple R-squared:  0.4234,    Adjusted R-squared:  0.4184
F-statistic: 85.39 on 17 and 1977 DF,  p-value: < 0.00000000000000022
```

```r
VaR_GARCH11 <- -(as.numeric(quantile(garch11_fit, probs = t_couv)))

plot(
  as.zoo(cbind(rend_dj,-VaR_GARCH11)),
  screens=1,
  col = c("black","lightseagreen"),
  main="VaR ARMA-GARCH(1)",
  xlab = "Temps",
  ylab = "Rendements"
)
points(as.zoo(rend_dj[as.numeric(rend_dj) < -VaR_GARCH11]), pch=20,
col="red")
legend("bottom",c("Rendements","VaR GARCH", "Dépassements de la VaR"),
         col = c("black","lightseagreen", 'red'),lty = c(1,1,NA),lwd =
         c(2,2,NA),pch = c(NA,NA,20))
```




```r
garch11_estimation_fit <-
  rugarch::ugarchfit(
    garch11_spec,
    rend_dj,
    out.sample = n_backtest
  )
```

```r
forecast_garch11 <-
  rugarch::ugarchforecast(
    garch11_estimation_fit,
    data = rend_dj,
    n.roll = n_backtest-1,
    n.ahead = 1
  )

VaR_GARCH_backtest <- -(as.numeric(quantile(forecast_garch11, probs = 0.01)))

test_VaR_GARCH<-
  rugarch::VaRTest(
    alpha = t_couv,
    actual = sample_backtest,
    VaR = -VaR_GARCH_backtest,
    conf.level = 0.95
  )

tibble(
  name = names(unlist(test_VaR_GARCH)),
  value = unlist(test_VaR_GARCH)
) |>
  print(n = 20)
```

```
# A tibble: 12 x 2
   name            value
   <chr>           <chr>
 1 expected.exceed 20
 2 actual.exceed   40
 3 uc.H0           Correct Exceedances
 4 uc.LRstat       15.654478305845
 5 uc.critical     3.84145882069412
 6 uc.LRp          0.0000760323231324689
 7 uc.Decision     Reject H0
 8 cc.H0           Correct Exceedances & Independent
 9 cc.LRstat       19.4369349243816
10 cc.critical     5.99146454710798
11 cc.LRp          0.00006016213067217
12 cc.Decision     Reject H0
```

```r
HIT <- if_else(sample_backtest < -VaR_GARCH_backtest, 1, 0)

HIT_centre <- HIT - 0.01

stats::Box.test(HIT_centre, lag = 5, type = "Ljung-Box")
```

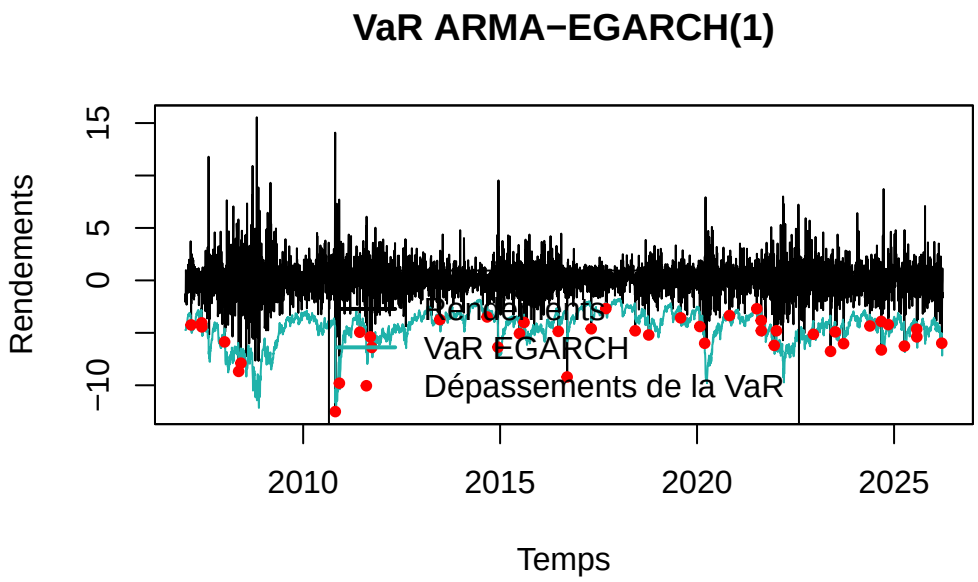
```
    Box-Ljung test

data:  HIT_centre
X-squared = 10.16, df = 5, p-value = 0.07084
```

```r
df <-
  tibble(
    HIT_centre = HIT_centre,
    rend = sample_backtest,
    rend2 = sample_backtest**2
  )

ARDL::ardl(
  HIT_centre ~ rend + rend2,
  data = df,
  order = 5
) |>
  summary()
```

```
Time series regression with "ts" data:
Start = 6, End = 2000

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
     Min       1Q   Median       3Q      Max
-0.56667 -0.02705  0.00565  0.02327  0.83890

Coefficients:
                   Estimate Std. Error t value             Pr(>|t|)
(Intercept)      -0.0023440  0.0034114  -0.687             0.492106
L(HIT_centre, 1)  0.0054403  0.0224528   0.242             0.808573
```

```
L(HIT_centre, 2) -0.0225418  0.0224088  -1.006             0.314569
L(HIT_centre, 3)  0.0727347  0.0223326   3.257             0.001145 **
L(HIT_centre, 4)  0.0508478  0.0223557   2.274             0.023044 *
L(HIT_centre, 5)  0.0269839  0.0222816   1.211             0.226025
rend             -0.0334506  0.0015165 -22.058 < 0.0000000000000002 ***
L(rend, 1)        0.0015558  0.0016936   0.919             0.358412
L(rend, 2)       -0.0014861  0.0016921  -0.878             0.379882
L(rend, 3)        0.0025751  0.0016938   1.520             0.128595
L(rend, 4)        0.0038172  0.0016953   2.252             0.024455 *
L(rend, 5)       -0.0021732  0.0016932  -1.283             0.199481
rend2             0.0112614  0.0004347  25.908 < 0.0000000000000002 ***
L(rend2, 1)      -0.0016795  0.0004999  -3.360             0.000796 ***
L(rend2, 2)      -0.0010498  0.0005002  -2.099             0.035974 *
L(rend2, 3)      -0.0014932  0.0004990  -2.993             0.002800 **
L(rend2, 4)      -0.0004462  0.0004993  -0.894             0.371626
L(rend2, 5)      -0.0020059  0.0004941  -4.059            0.0000511 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1119 on 1977 degrees of freedom
Multiple R-squared:  0.3688,    Adjusted R-squared:  0.3633
F-statistic: 67.94 on 17 and 1977 DF,  p-value: < 0.00000000000000022
```

```r
VaR_EGARCH11 <- -(as.numeric(quantile(egarch11_fit, probs = t_couv)))

plot(
  as.zoo(cbind(rend_dj,-VaR_EGARCH11)),
  screens=1,
  col = c("black","lightseagreen"),
  main="VaR ARMA-EGARCH(1)",
  xlab = "Temps",
  ylab = "Rendements"
)
points(as.zoo(rend_dj[as.numeric(rend_dj) < -VaR_EGARCH11]), pch=20,
col="red")
legend("bottom",c("Rendements","VaR EGARCH", "Dépassements de la VaR"),
         col = c("black","lightseagreen", 'red'),lty = c(1,1,NA),lwd =
         c(2,2,NA),pch = c(NA,NA,20))
```




```
egarch11_estimation_fit <-
  rugarch::ugarchfit(
    egarch11_spec,
    rend_dj,
    out.sample = n_backtest
  )

forecast_egarch11 <-
  rugarch::ugarchforecast(
    egarch11_estimation_fit,
    data = rend_dj,
    n.roll = n_backtest-1,
    n.ahead = 1
  )

VaR_EGARCH_backtest <- -(as.numeric(quantile(forecast_egarch11, probs =
0.01)))

test_VaR_EGARCH<-
  rugarch::VaRTest(
    alpha = t_couv,
    actual = sample_backtest,
    VaR = -VaR_EGARCH_backtest,
```

```r
    conf.level = 0.95
  )

tibble(
  name = names(unlist(test_VaR_EGARCH)),
  value = unlist(test_VaR_EGARCH)
) |>
  print(n = 20)
```

```
# A tibble: 12 x 2
   name            value
   <chr>           <chr>
 1 expected.exceed 20
 2 actual.exceed   22
 3 uc.H0           Correct Exceedances
 4 uc.LRstat       0.195668793956935
 5 uc.critical     3.84145882069412
 6 uc.LRp          0.658239821679998
 7 uc.Decision     Fail to Reject H0
 8 cc.H0           Correct Exceedances & Independent
 9 cc.LRstat       10.5160839686686
10 cc.critical     5.99146454710798
11 cc.LRp          0.00520548717218117
12 cc.Decision     Reject H0
```

```r
HIT <- if_else(sample_backtest < -VaR_EGARCH_backtest, 1, 0)

HIT_centre <- HIT - 0.01

stats::Box.test(HIT_centre, lag = 5, type = "Ljung-Box")
```

```
    Box-Ljung test

data:  HIT_centre
X-squared = 33.176, df = 5, p-value = 0.000003472
```

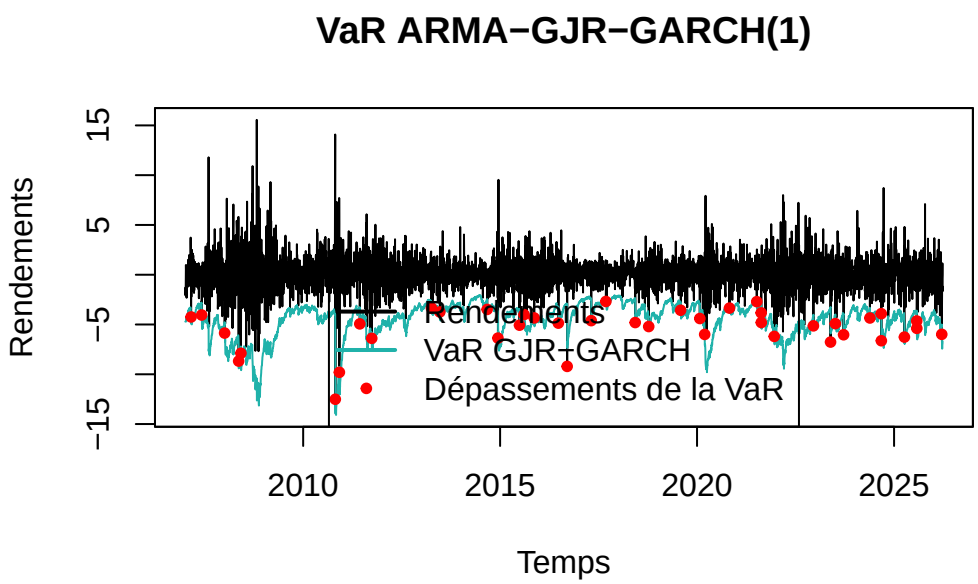
```r
df <-
  tibble(
    HIT_centre = HIT_centre,
    rend = sample_backtest,
    rend2 = sample_backtest**2
```

```r
  )

ARDL::ardl(
  HIT_centre ~ rend + rend2,
  data = df,
  order = 5
) |>
  summary()
```

```
Time series regression with "ts" data:
Start = 6, End = 2000

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
     Min       1Q   Median       3Q      Max
-0.39452 -0.01480  0.00553  0.01578  0.88532

Coefficients:
                   Estimate Std. Error t value             Pr(>|t|)
(Intercept)      -0.0076745  0.0027125  -2.829             0.004711 **
L(HIT_centre, 1)  0.1129868  0.0224690   5.029          0.000000539 ***
L(HIT_centre, 2) -0.0412201  0.0226192  -1.822             0.068554 .
L(HIT_centre, 3)  0.0295952  0.0226606   1.306             0.191698
L(HIT_centre, 4)  0.0014550  0.0226525   0.064             0.948791
L(HIT_centre, 5)  0.0145685  0.0224671   0.648             0.516776
rend             -0.0205429  0.0011903 -17.258 < 0.0000000000000002 ***
L(rend, 1)        0.0034156  0.0012777   2.673             0.007575 **
L(rend, 2)       -0.0004147  0.0012790  -0.324             0.745785
L(rend, 3)       -0.0001827  0.0012805  -0.143             0.886537
L(rend, 4)        0.0017568  0.0012793   1.373             0.169819
L(rend, 5)       -0.0013831  0.0012805  -1.080             0.280215
rend2             0.0077204  0.0003421  22.567 < 0.0000000000000002 ***
L(rend2, 1)      -0.0016319  0.0003816  -4.276          0.000019922 ***
L(rend2, 2)      -0.0004869  0.0003831  -1.271             0.203874
L(rend2, 3)      -0.0005263  0.0003823  -1.376             0.168828
L(rend2, 4)      -0.0003430  0.0003823  -0.897             0.369815
L(rend2, 5)      -0.0012804  0.0003796  -3.373             0.000757 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.08781 on 1977 degrees of freedom
Multiple R-squared:  0.2993,    Adjusted R-squared:  0.2933
F-statistic: 49.68 on 17 and 1977 DF,  p-value: < 0.00000000000000022
```

```
VaR_GJR_GARCH11 <- -(as.numeric(quantile(gjr_garch11_fit, probs = t_couv)))

plot(
  as.zoo(cbind(rend_dj,-VaR_GJR_GARCH11)),
  screens=1,
  col = c("black","lightseagreen"),
  main="VaR ARMA-GJR-GARCH(1)",
  xlab = "Temps",
  ylab = "Rendements"
)
points(as.zoo(rend_dj[as.numeric(rend_dj) < -VaR_GJR_GARCH11]), pch=20,
col="red")
legend("bottom",c("Rendements","VaR GJR-GARCH", "Dépassements de la VaR"),
         col = c("black","lightseagreen", 'red'),lty = c(1,1,NA),lwd =
         c(2,2,NA),pch = c(NA,NA,20))
```



```r
gjr_garch11_estimation_fit <-
  rugarch::ugarchfit(
    gjr_garch11_spec,
    rend_dj,
    out.sample = n_backtest
  )

forecast_gjr_garch11 <-
  rugarch::ugarchforecast(
    gjr_garch11_estimation_fit,
    data = rend_dj,
    n.roll = n_backtest-1,
    n.ahead = 1
  )

VaR_GJR_GARCH_backtest <- -(as.numeric(quantile(forecast_gjr_garch11, probs =
0.01)))

test_VaR_GJR_GARCH<-
  rugarch::VaRTest(
    alpha = t_couv,
    actual = sample_backtest,
    VaR = -VaR_GJR_GARCH_backtest,
    conf.level = 0.95
  )

tibble(
  name = names(unlist(test_VaR_GJR_GARCH)),
  value = unlist(test_VaR_GJR_GARCH)
) |>
  print(n = 20)
```

```
# A tibble: 12 x 2
   name            value
   <chr>           <chr>
 1 expected.exceed 20
 2 actual.exceed   24
 3 uc.H0           Correct Exceedances
 4 uc.LRstat       0.759520981315486
 5 uc.critical     3.84145882069412
 6 uc.LRp          0.38347847189258
 7 uc.Decision     Fail to Reject H0
 8 cc.H0           Correct Exceedances & Independent
```

```
 9 cc.LRstat       10.0424119644939
10 cc.critical     5.99146454710798
11 cc.LRp          0.00659656656821295
12 cc.Decision     Reject H0
```

```r
HIT <- if_else(sample_backtest < -VaR_GJR_GARCH_backtest, 1, 0)

HIT_centre <- HIT - 0.01

stats::Box.test(HIT_centre, lag = 5, type = "Ljung-Box")
```

```
    Box-Ljung test

data:  HIT_centre
X-squared = 28.895, df = 5, p-value = 0.00002431
```

```r
df <-
  tibble(
    HIT_centre = HIT_centre,
    rend = sample_backtest,
    rend2 = sample_backtest**2
  )

ARDL::ardl(
  HIT_centre ~ rend + rend2,
  data = df,
  order = 5
) |>
  summary()
```

```
Time series regression with "ts" data:
Start = 6, End = 2000

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
     Min       1Q   Median       3Q      Max
-0.41614 -0.01672  0.00582  0.01706  0.87569
```

```
Coefficients:
                   Estimate Std. Error t value             Pr(>|t|)
(Intercept)      -0.0072480  0.0028129  -2.577             0.010046 *
L(HIT_centre, 1)  0.0850244  0.0225091   3.777             0.000163 ***
L(HIT_centre, 2)  0.0058460  0.0226194   0.258             0.796087
L(HIT_centre, 3)  0.0233207  0.0226570   1.029             0.303467
L(HIT_centre, 4)  0.0055414  0.0226505   0.245             0.806755
L(HIT_centre, 5)  0.0035904  0.0225159   0.159             0.873323
rend             -0.0220566  0.0012373 -17.826 < 0.0000000000000002 ***
L(rend, 1)        0.0035582  0.0013340   2.667             0.007707 **
L(rend, 2)        0.0007856  0.0013352   0.588             0.556331
L(rend, 3)       -0.0002775  0.0013359  -0.208             0.835438
L(rend, 4)        0.0012833  0.0013348   0.961             0.336478
L(rend, 5)       -0.0023595  0.0013344  -1.768             0.077188 .
rend2             0.0082472  0.0003546  23.257 < 0.0000000000000002 ***
L(rend2, 1)      -0.0016310  0.0003980  -4.098            0.0000433 ***
L(rend2, 2)      -0.0009898  0.0003989  -2.481             0.013180 *
L(rend2, 3)      -0.0005334  0.0003988  -1.338             0.181202
L(rend2, 4)      -0.0003598  0.0003988  -0.902             0.367070
L(rend2, 5)      -0.0010857  0.0003957  -2.744             0.006124 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09102 on 1977 degrees of freedom
Multiple R-squared:  0.3092,    Adjusted R-squared:  0.3033
F-statistic: 52.06 on 17 and 1977 DF,  p-value: < 0.00000000000000022
```

```r
df_VaR <-
  tibble(
    VaR_VCov = VaR_VCov,
    VaR_GARCH11 = VaR_GARCH11,
    VaR_EGARCH11 = VaR_EGARCH11,
    VaR_GJR_GARCH11 = VaR_GJR_GARCH11
  ) |>
  drop_na()

df_VaR_long <-
  df_VaR |>
  mutate(temps = row_number()) |>
  pivot_longer(
    cols = -temps,
    names_to = "model",
    values_to = "VaR"
```

```r
  ) |>
  mutate(
    .by = temps,
    moyenne_VaR = mean(VaR)
  ) |>
  mutate(
    ecart = (VaR - moyenne_VaR) / moyenne_VaR
  ) |>
  summarize(
    .by = model,
    MRB = abs(mean(ecart))
  ) |>
  print()
```

```
# A tibble: 4 x 2
  model              MRB
  <chr>            <dbl>
1 VaR_VCov        0.0106
2 VaR_GARCH11     0.100
3 VaR_EGARCH11    0.0406
4 VaR_GJR_GARCH11 0.0491
```